

Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: DispM-mwRev-Ch.sil etc.

Creation date: 9-Nov-82 0:01:43

For: Spruce

33 total sheets = 32 pages, 1 copy.

Problems encountered:

Font HELVETICA24B substituted for font HELVE24.

Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

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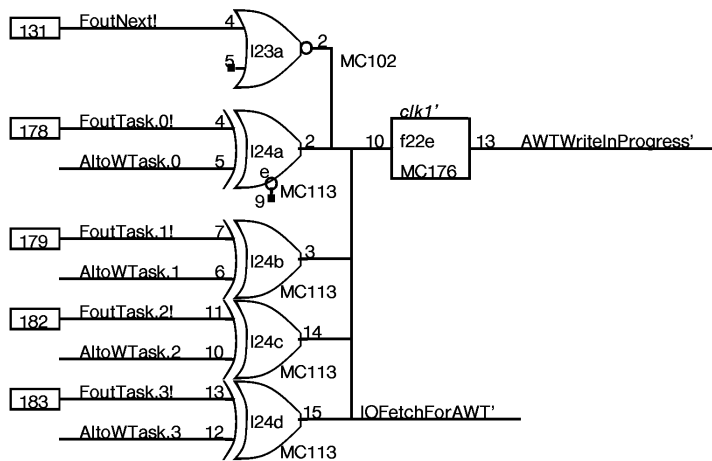
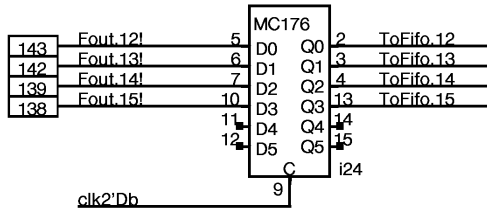
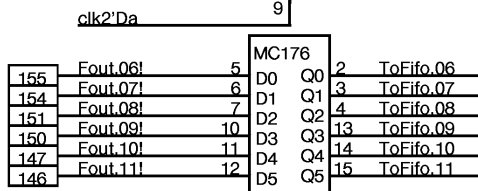
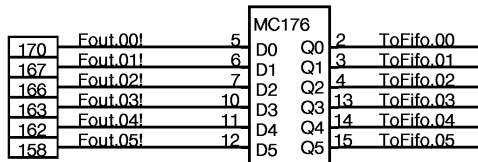
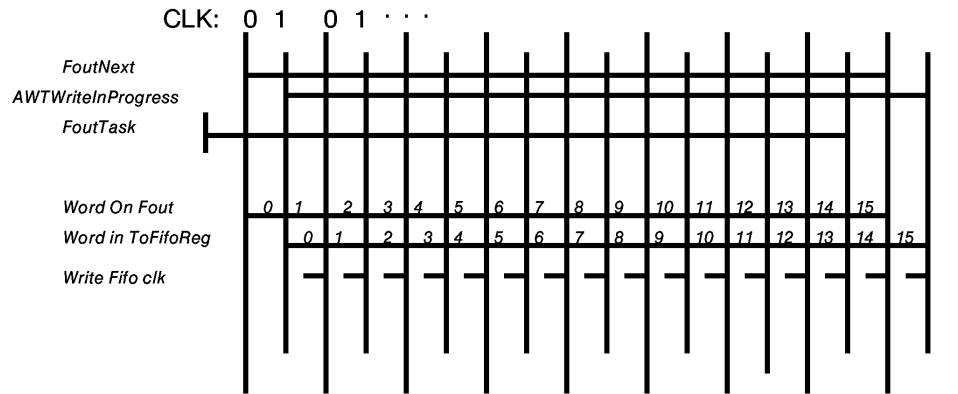
Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

DORADO SCHEMATICS

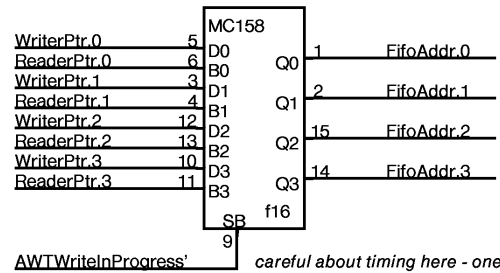
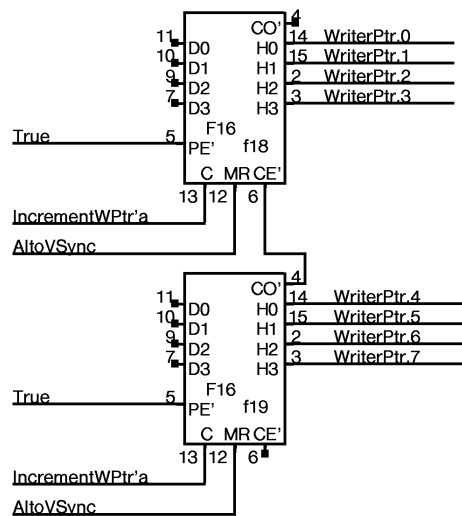
Display M

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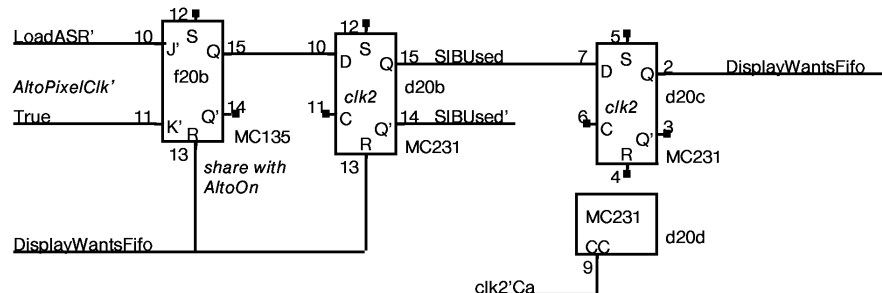
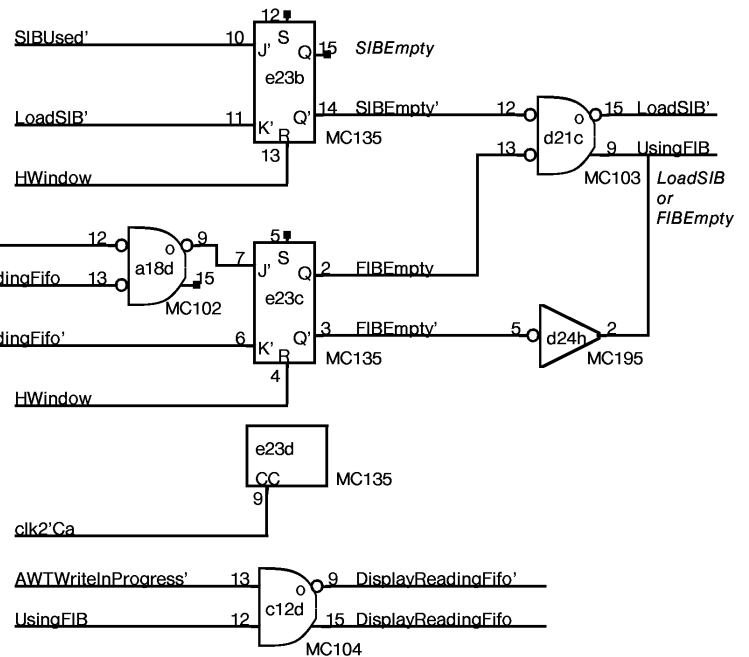
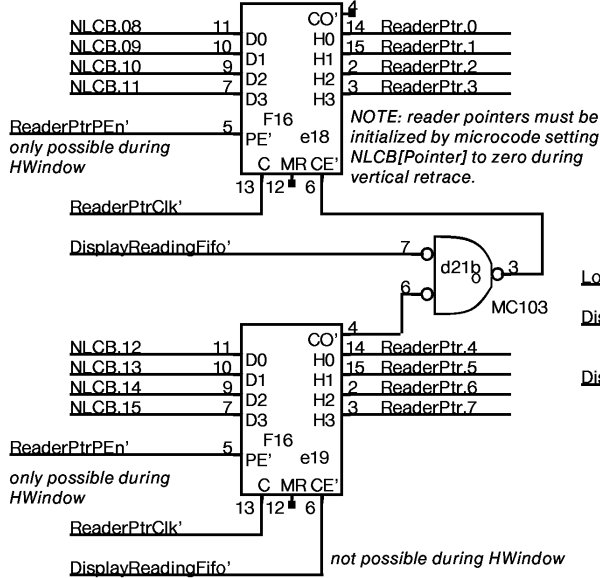
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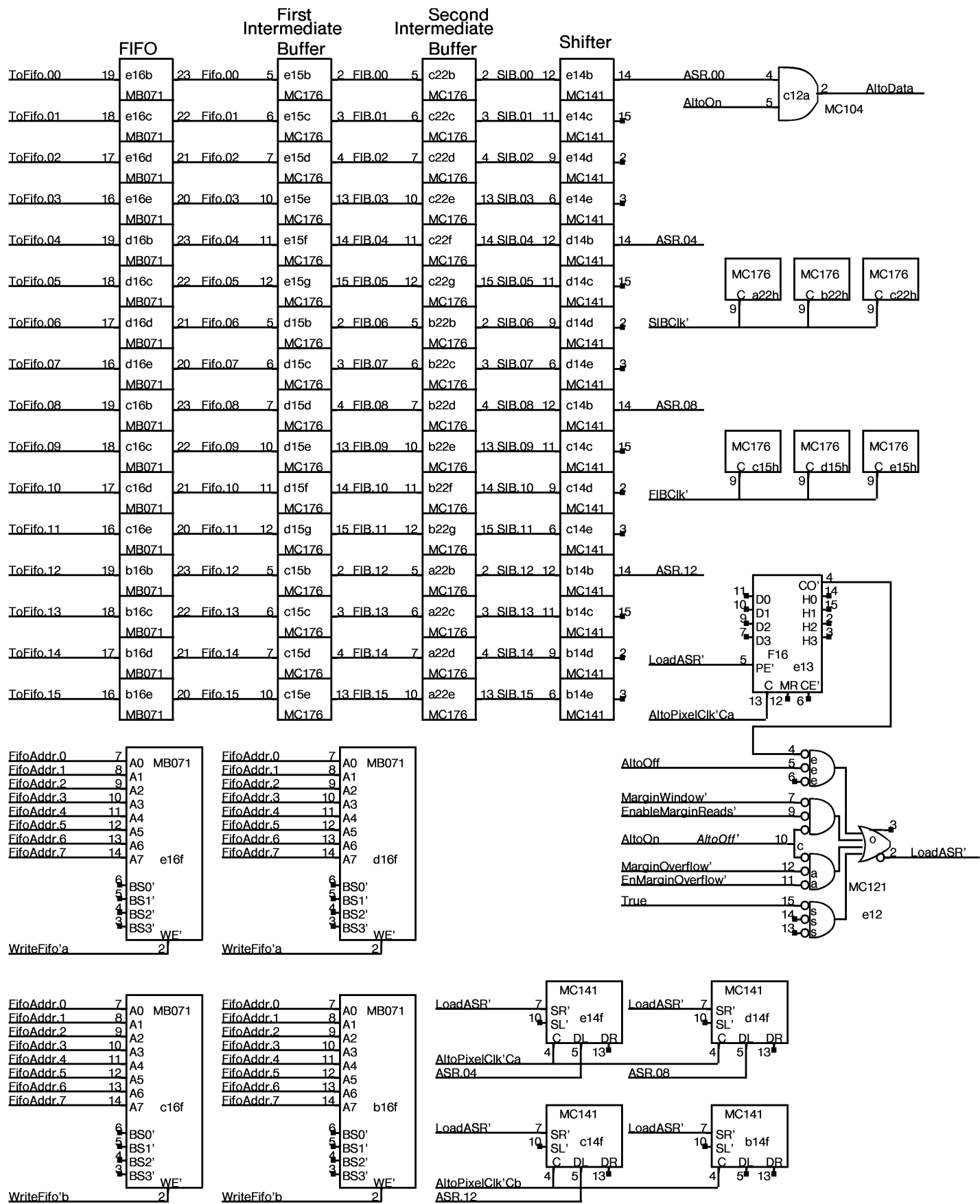


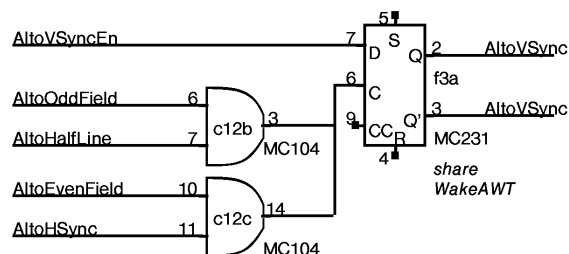
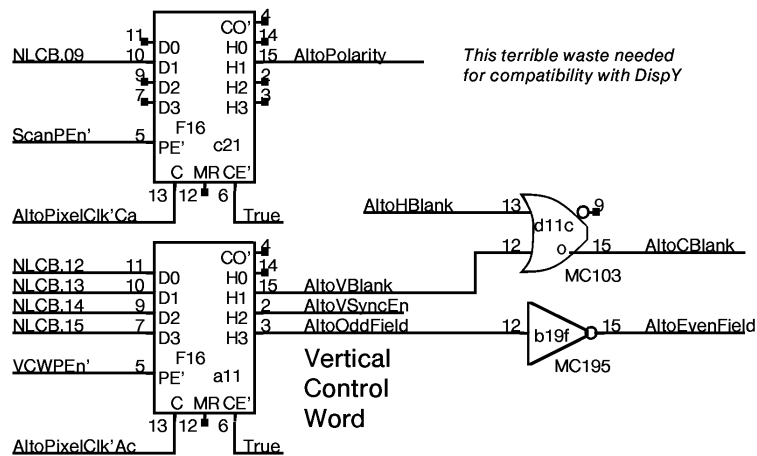
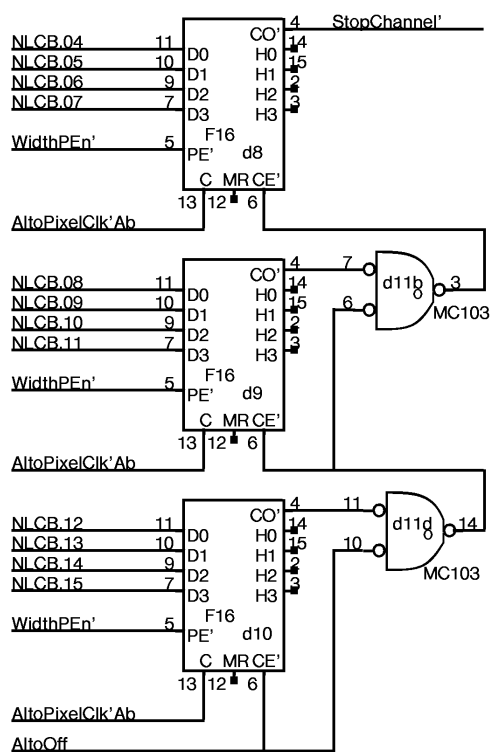
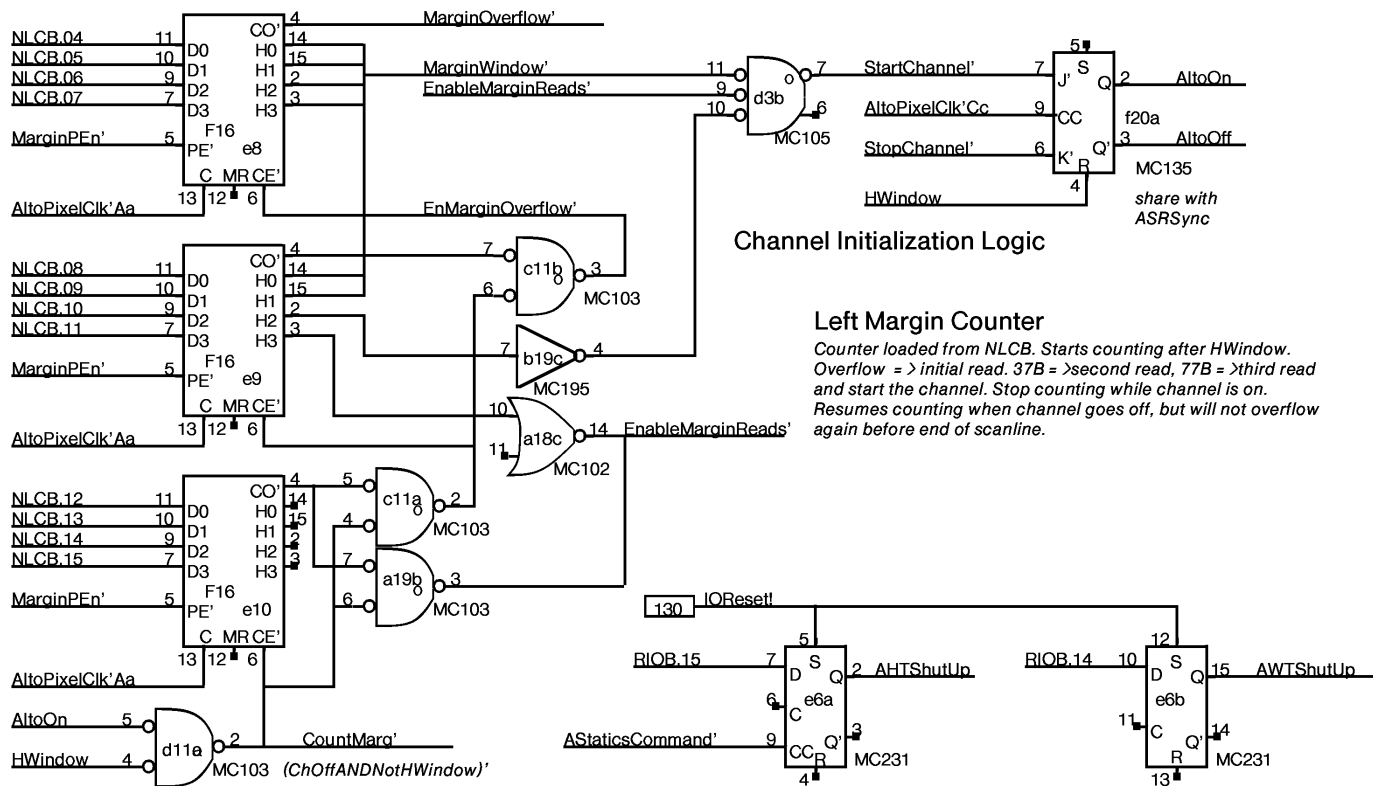
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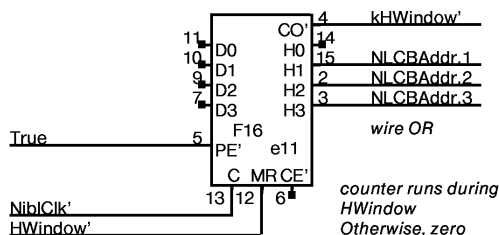
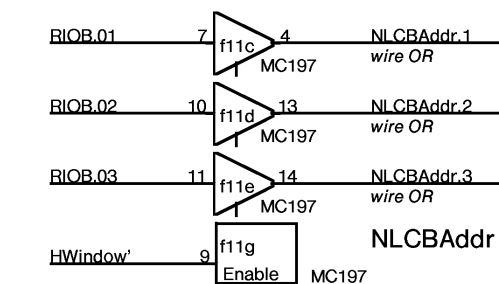


Reader Pointer



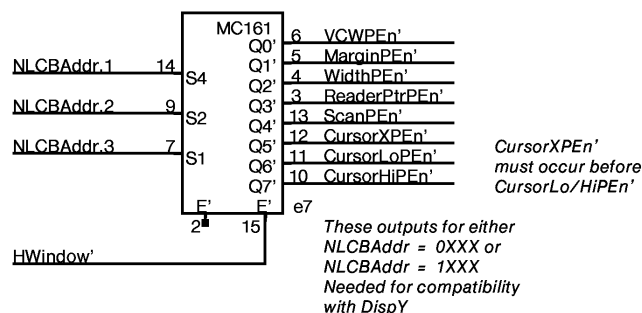






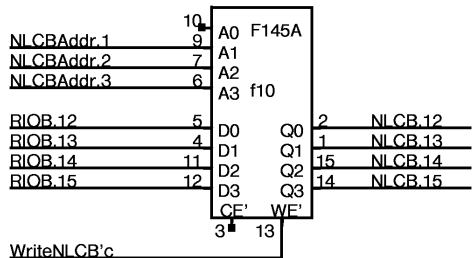
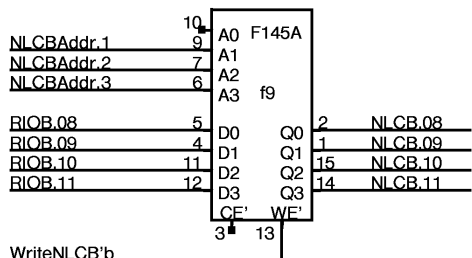
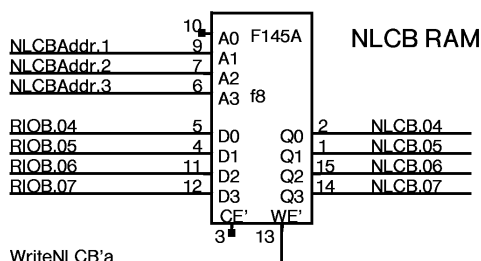
Note: rate is Pixel/4, unlike
DispY which is Pixel/2.
Microcode compensates.

CLCB load enables

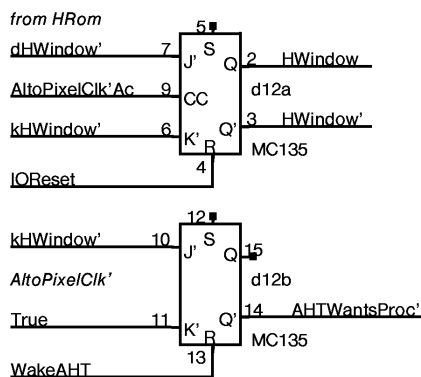


NOTE:

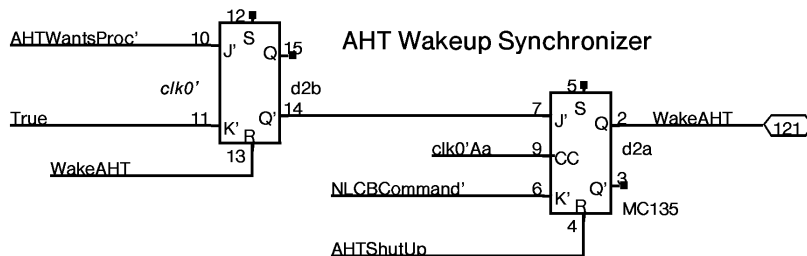
There are only 8 entries in NLCB.
The MSB (NLCBAddr.0) is ignored.
Outputs to 0XXX or 1XXX write
words 0XXX of NLCB. During HWindow,
CLCB entries are redundantly written
twice, first when counter goes 00-07
and again when counter goes 10-17.



HWindow

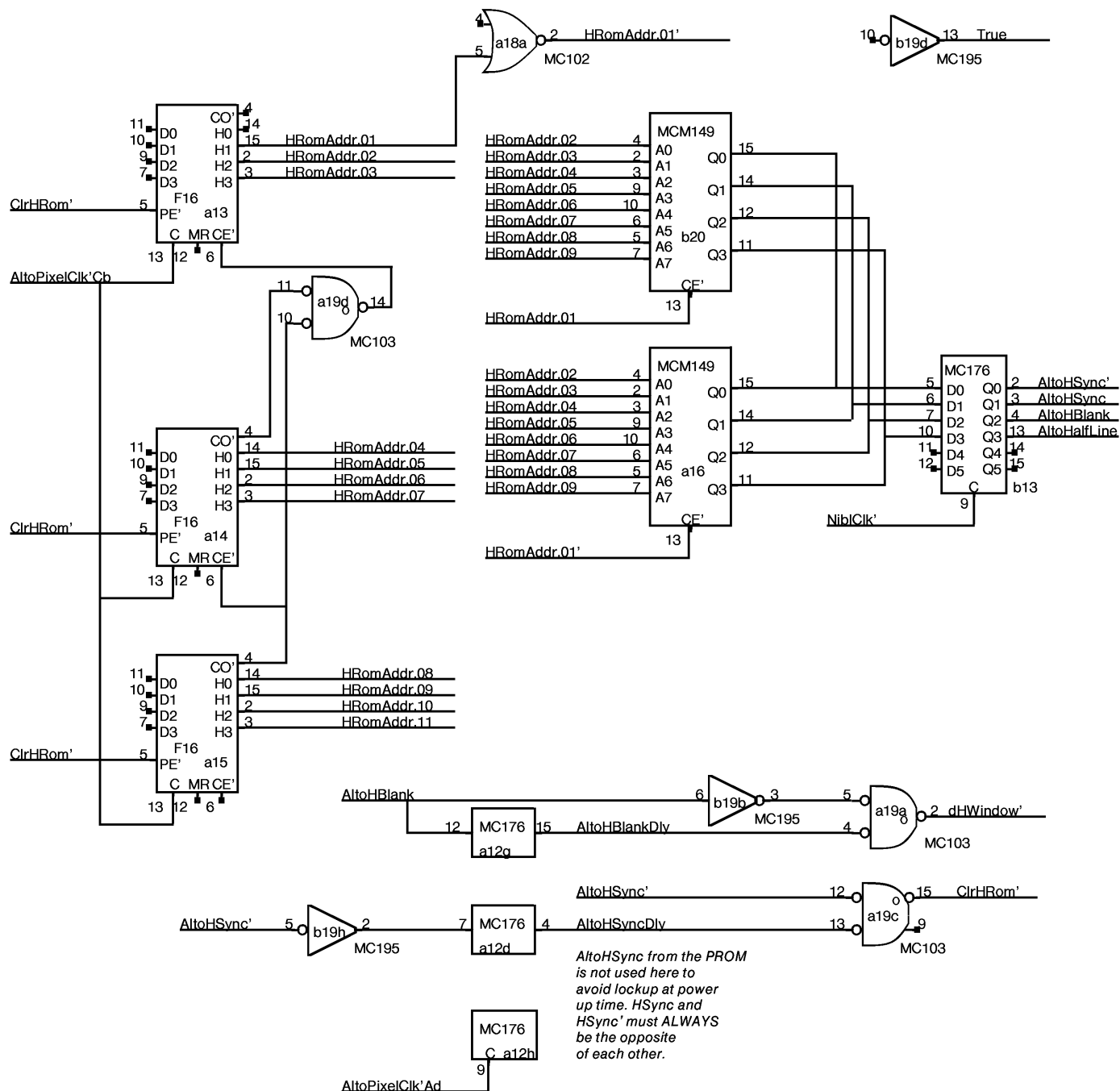


AHT Wakeup Synchronizer



For convenience, use NLCB command
to kill wakeup. Assumes AHT will always
do some NLCB command whenever it is
awakened. Default would be to redundantly
load NLCB[0] with same data.

requires minimum 4 instruction loop



NEXT Word Control Block Flag (NextWCBFlag)
Current Word Control Block Flag (CurrentWCBFlag)

A Word Control Block is a pair of values called
Address and MunchCount, for either the CURRENT
or the NEXT scanline.

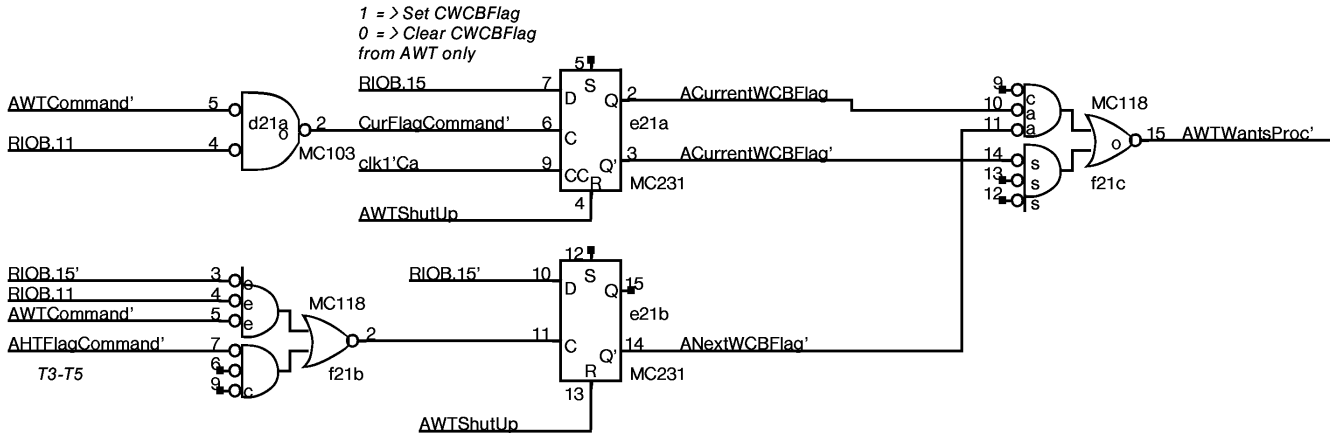
For AWT Commands:		For AHT commands:	
RIOB	Means	RIOB	Means
= 1c	Set CWCBCFlag and Clear NWCBCFlag	= 0c	Set ANextWCBFlag
= 0c	Clear CWCBCFlag		

WakeUp conditions:

WakeAHT: at end of every HWindow

WakeAWT: (CurrentWCBFlag) OR (NextWCBFlag AND NOT(CurrentWCBFlag))

Flag management	SET	CLEARED
NextWCBFlag	by AHT when it has filled the NextWCB	by AWT when it has copied NextWCB into CurrentWCB
CurrentWCBFlag	by AWT when it has copied NextWCB into CurrentWCB	by AWT when it has sent out all the data from the CurrentWCB



Whatever	1	P1 SPARE
	16	P16
	8	P8 a17

Whatever	1	P1 SPARE
	16	P16
	8	P8 b24

Whatever	1	P1 SPARE
	16	P16
	8	P8 d22

Whatever	1	P1 SPARE
	16	P16
	8	P8 e4

Whatever	1	P1 SPARE
	16	P16
	8	P8 a20

Whatever	1	P1 SPARE
	16	P16
	8	P8 c1

Whatever	1	P1 SPARE
	16	P16
	8	P8 c2

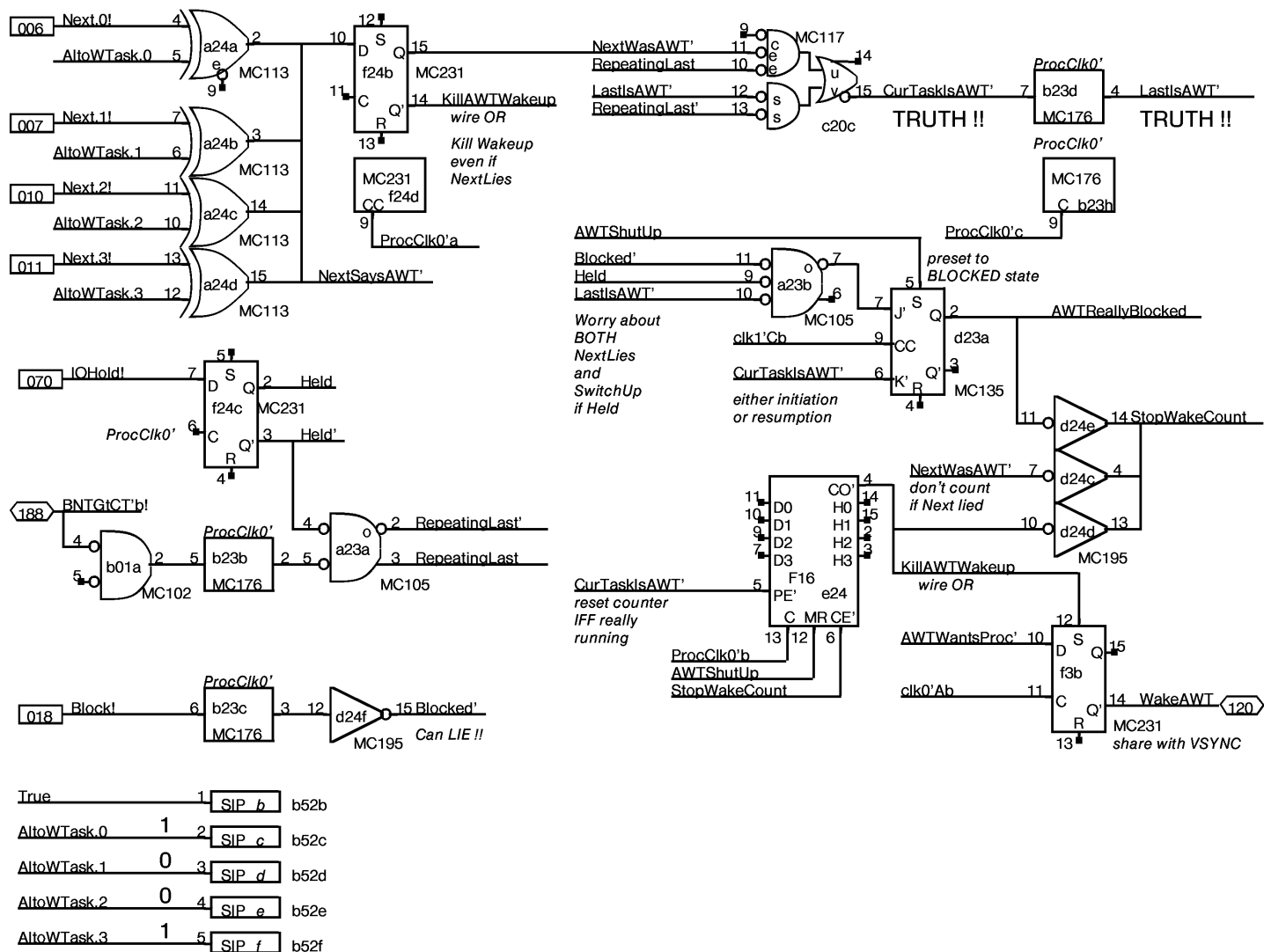
Whatever	1	P1 SPARE
	16	P16
	8	P8 e5

Whatever	1	P1 SPARE
	16	P16
	8	P8 j4

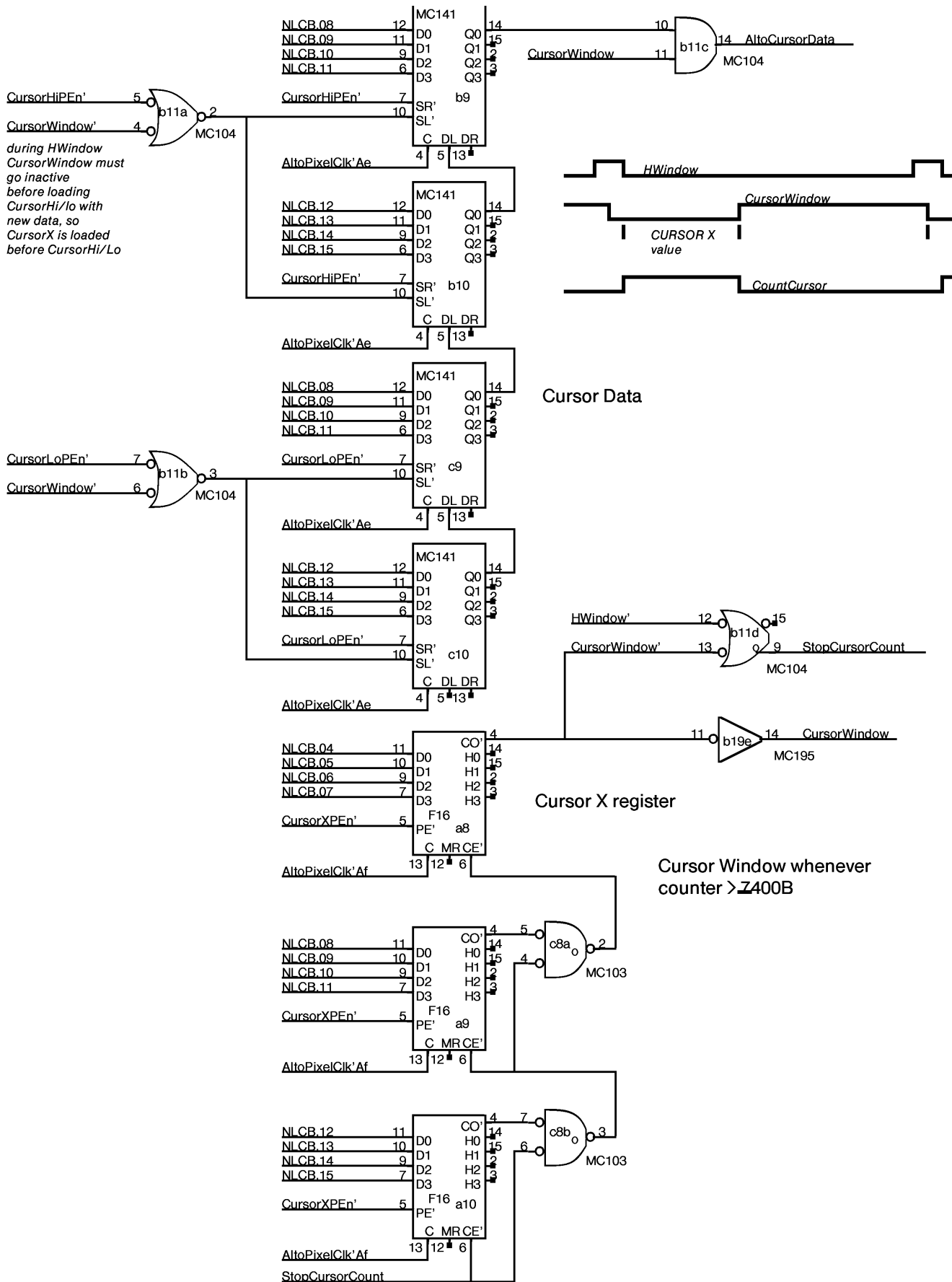
Whatever	1	P1 SPARE
	16	P16
	8	P8 j5

Whatever	1	P1 SPARE
	16	P16
	8	P8 k23

Whatever	1	P1 SPARE
	16	P16
	8	P8 l3



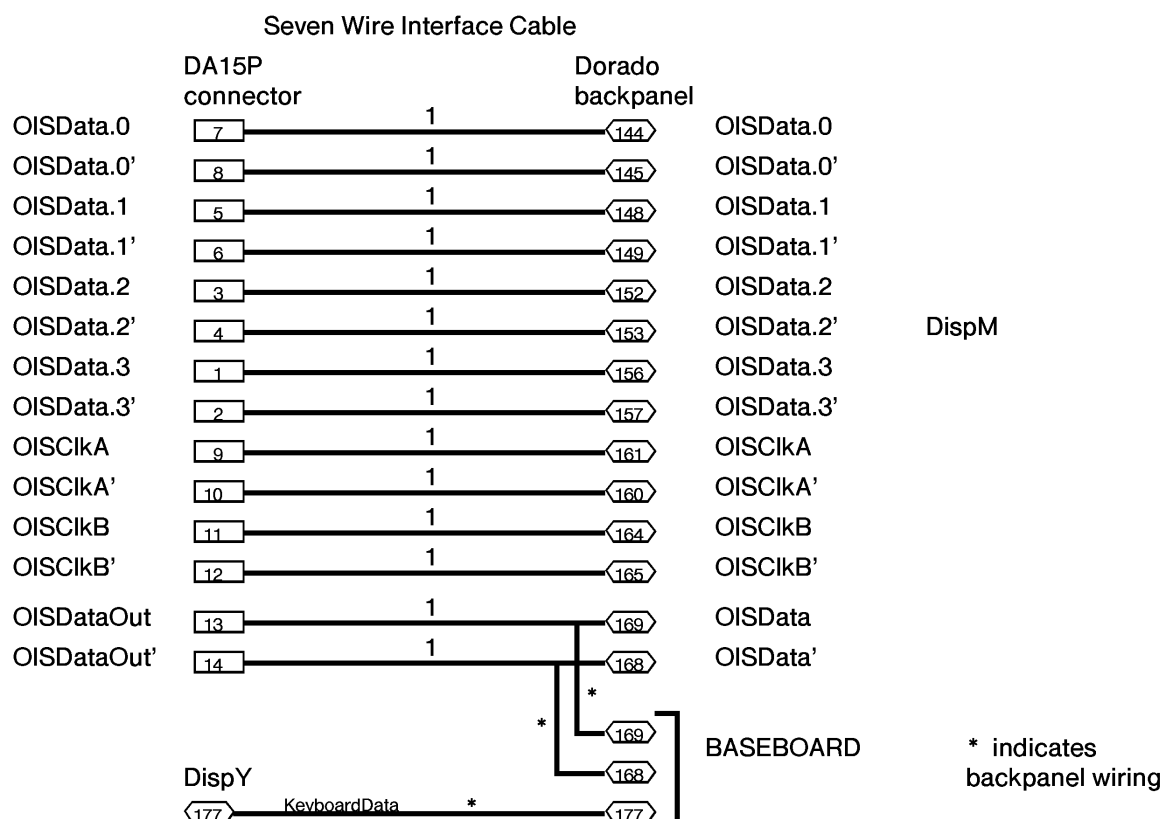
For AWT Task = 9D = 11B
cut legs 3 and 4

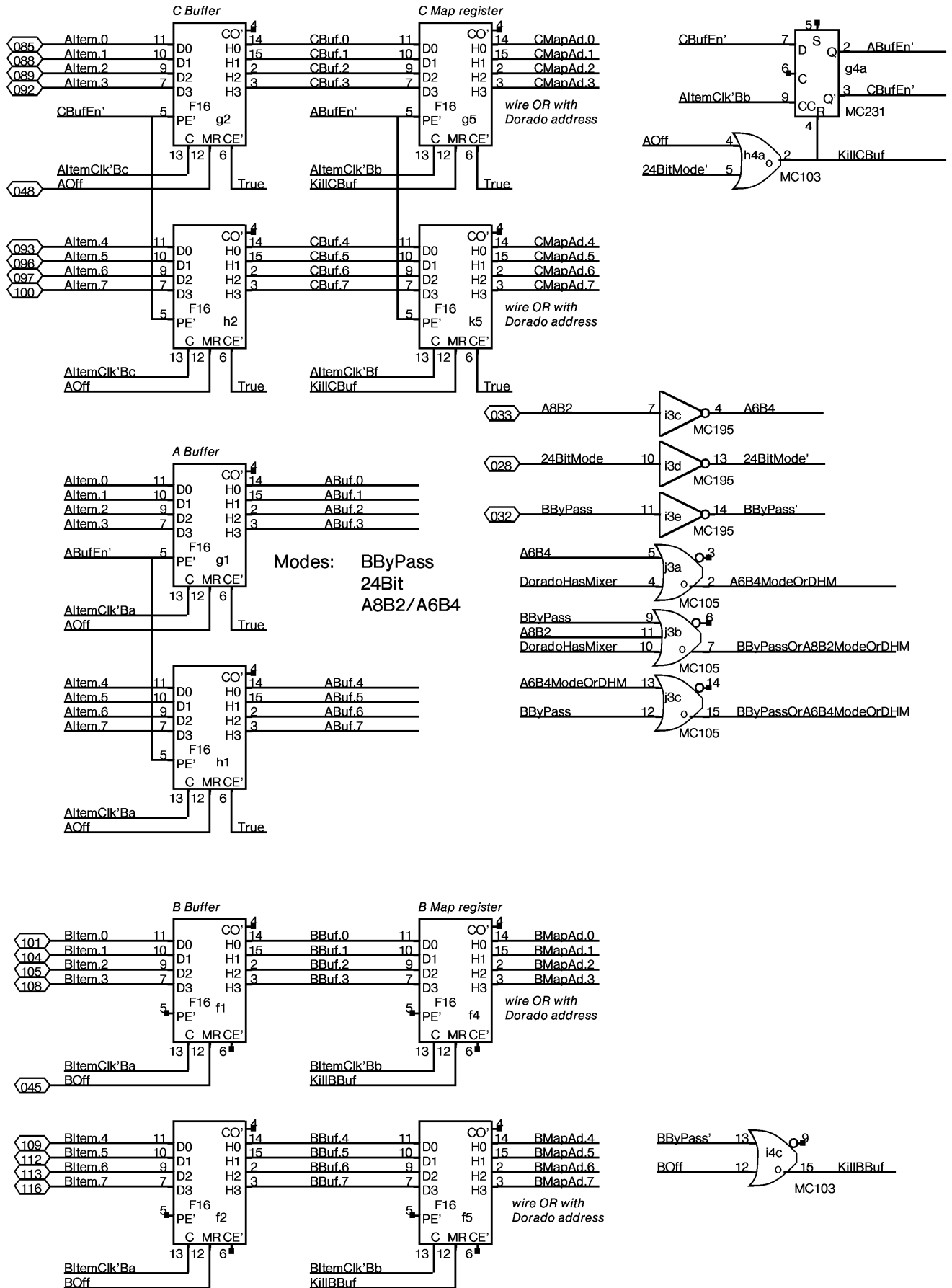




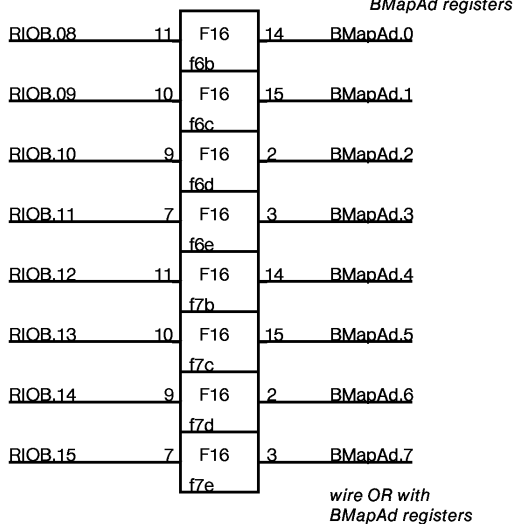
Alto Display Controller
Next Line Control Block Format

Address	Name	Format
0	VCW vertical control word	0,..0, VBlank, VSync, OddField
1	Margin	LMarg[00..11]
2	Width	Width[00..11]
3	FifoAddr	FifoAddr[0..7] *must be even
4	Scan	AltoPolarity,0,0,0,0,0,0,
15	CursorX	CursorXCount[0..11]
16	CursorLo	CursorLoByte[4..11]
17	CursorHi	CursorHiByte[4..11]

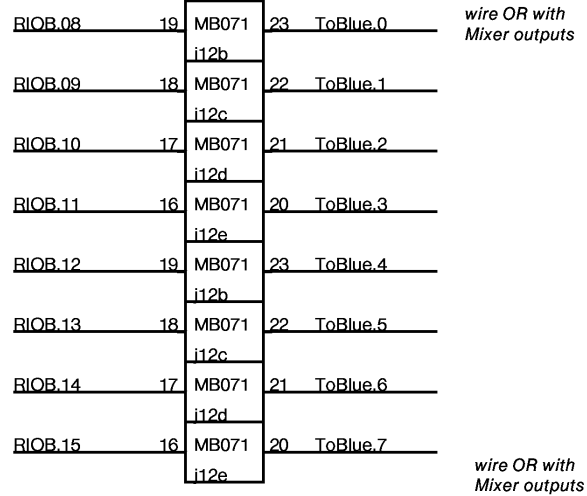




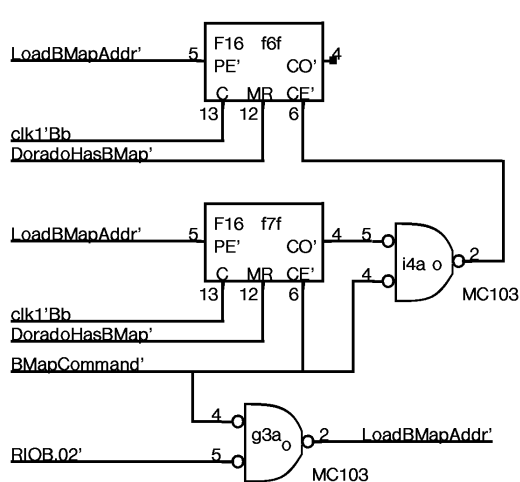
Dorado BMap address



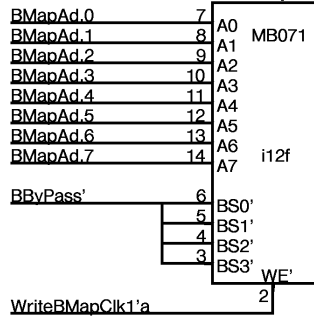
B Map



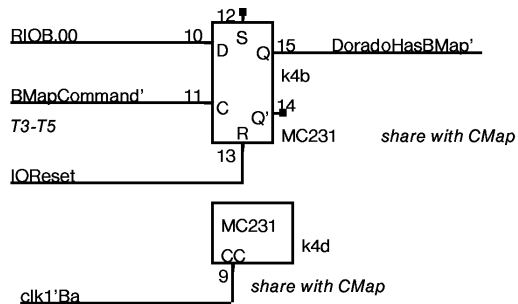
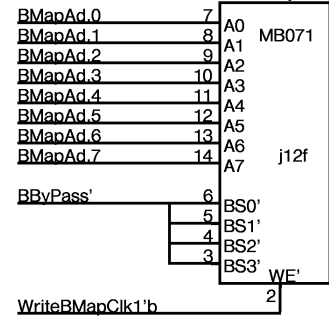
Dorado BMap address



B Map



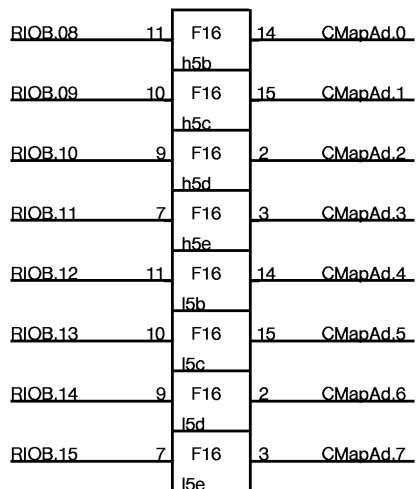
B Map



NOTE: in order to read/write BMap, BByPass mode must be ON and the B Channel must be OFF. This is inconvenient, but it does the right thing for real time mode switching.

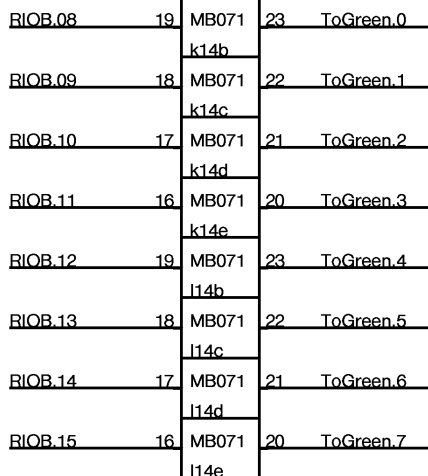
Dorado CMap address

wire OR with
CMapAd registers



wire OR with
CMapAd registers

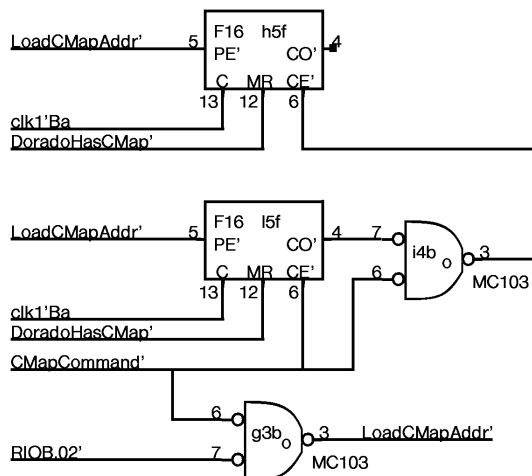
C Map



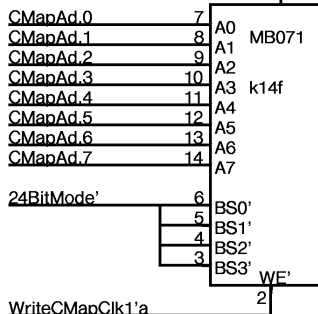
wire OR with
Mixer outputs

wire OR with
Mixer outputs

Dorado CMap address

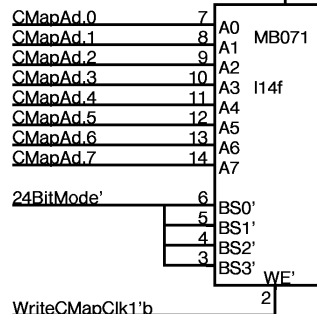


C Map

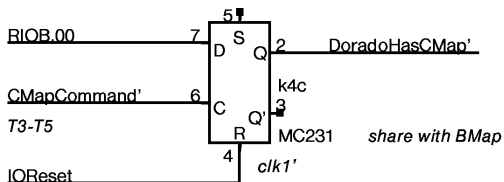


24BitMode' 6 BS0' 5 BS1' 4 BS2' 3 BS3' WE' 2 WriteCMapClk1'a

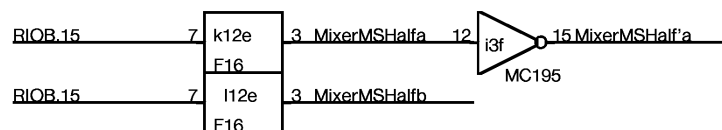
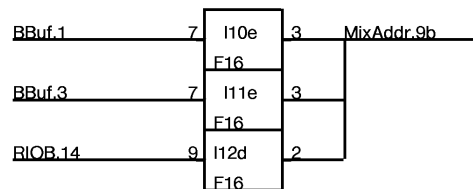
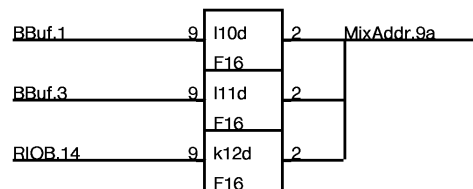
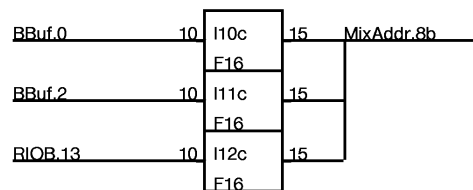
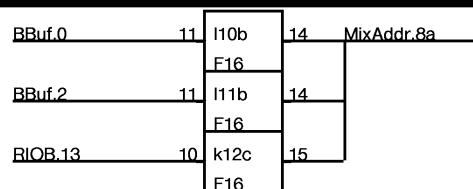
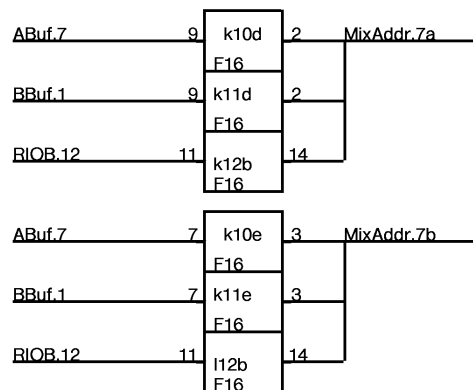
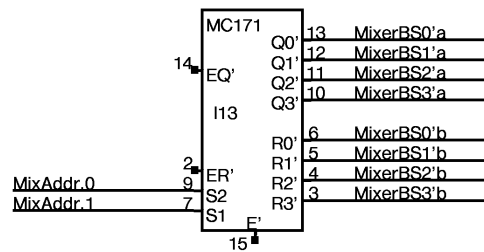
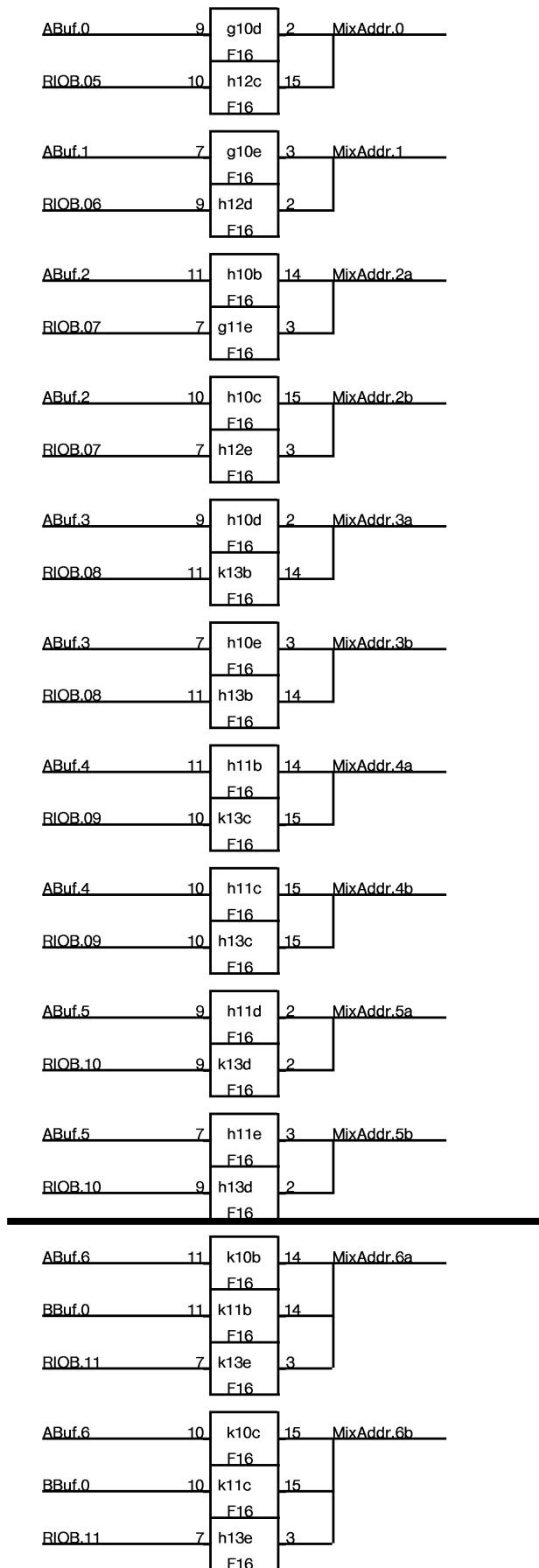
C Map



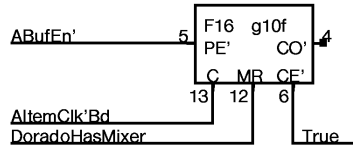
24BitMode' 6 BS0' 5 BS1' 4 BS2' 3 BS3' WE' 2 WriteCMapClk1'b



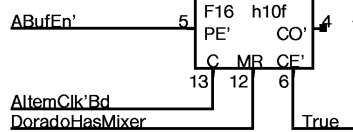
NOTE: in order to read/write CMap, 24BitMode must be ON and The A channel must be OFF. This is inconvenient, but it does the right thing for real time mode switching.



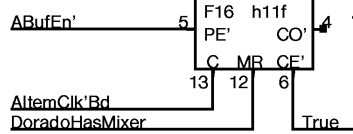
A Spare, Spare, 0, 1



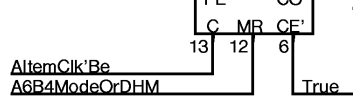
B 2a, 2b, 3a, 3b



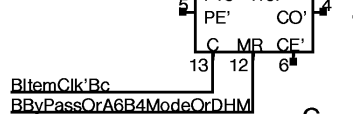
C 4a, 4b, 5a, 5b



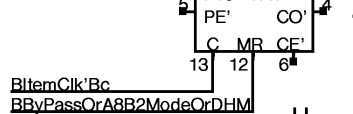
D 6a, 6b, 7a, 7b: Chan A



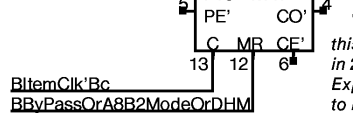
F 8a, 8b, 9a, 9b: Chan B



G 8a, 8b, 9a, 9b: Chan B

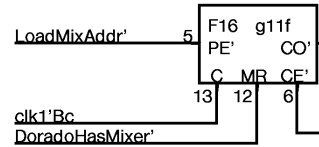


H 6a, 6b, 7a, 7b: Chan B

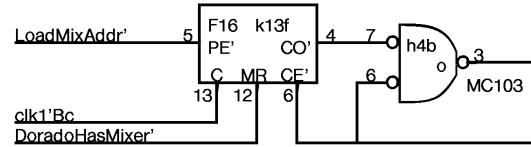


this must be OFF in 24Bit mode. Expect BByPass and A8B2 to be set whenever 24Bit mode is set.

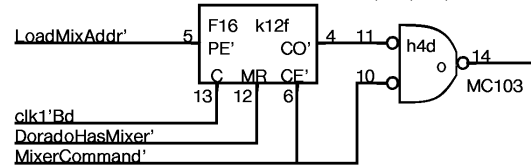
U , , , 2a



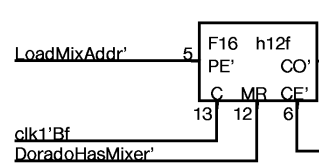
W 3a, 4a, 5a, 6a



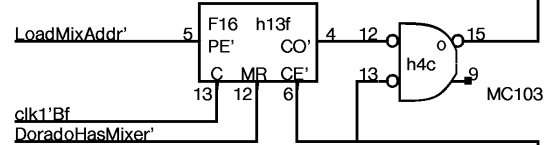
Y 7a, 8a, 9a, MSa



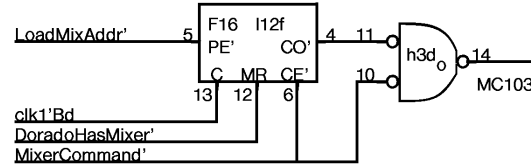
V Spare, 0, 1, 2b



X 3b, 4b, 5b, 6b

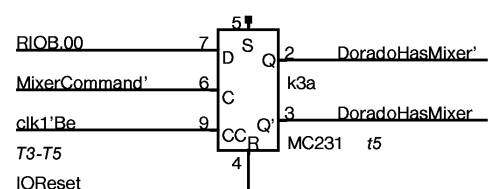
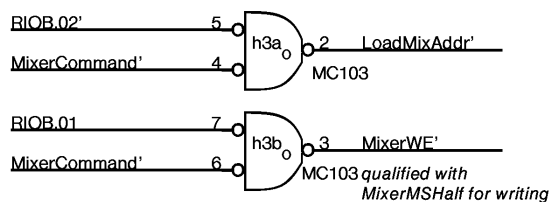


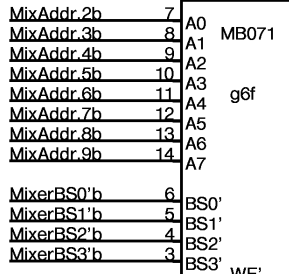
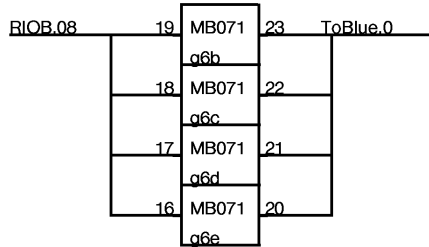
Z 7b, 8b, 9b, MSb



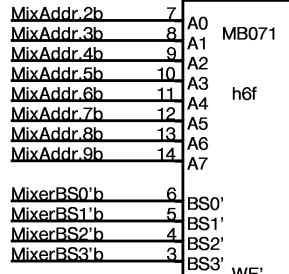
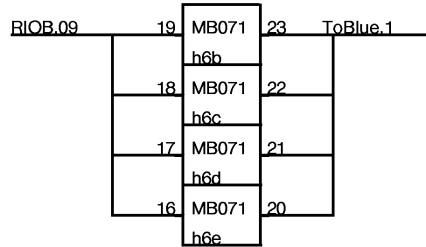
NOTE: B Channel must be started by microcode one pixel clk tick earlier than A channel for 24BitMode to align properly.

NOTE: Pixel clock must run at 2X rate for 24BitMode to work across entire screen !!

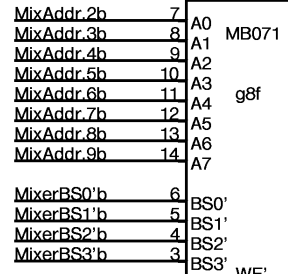
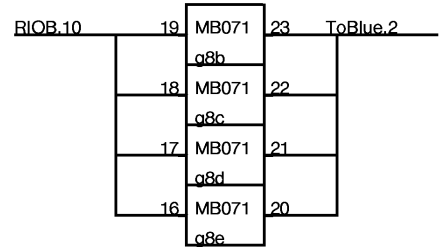




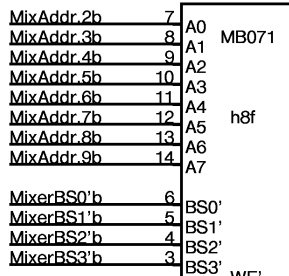
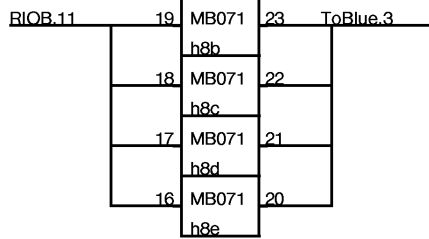
WriteMixerLo'Bb 2



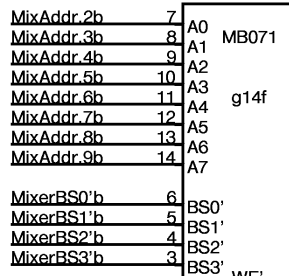
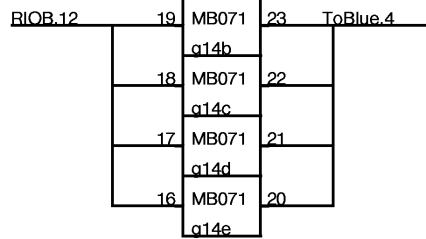
WriteMixerLo'Bb 2



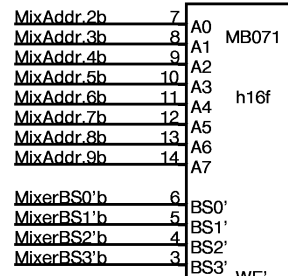
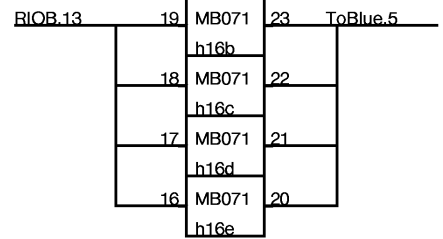
WriteMixerLo'Bc 2



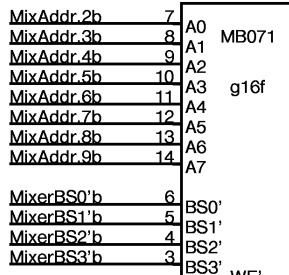
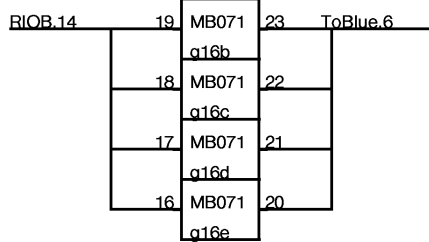
WriteMixerLo'Bc 2



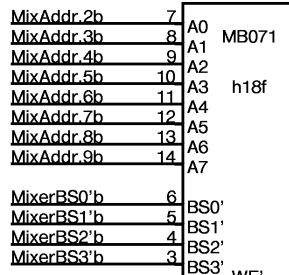
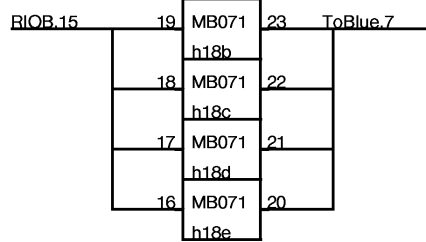
WriteMixerLo'Dc 2



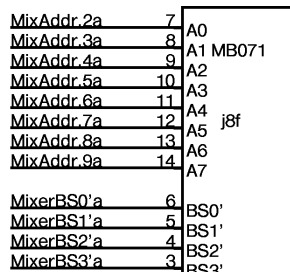
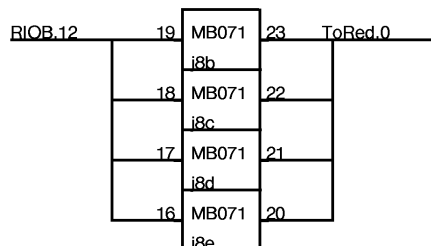
WriteMixerLo'Db 2



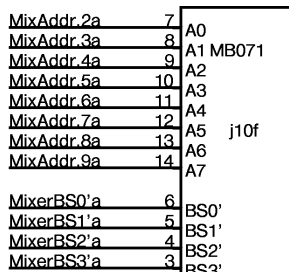
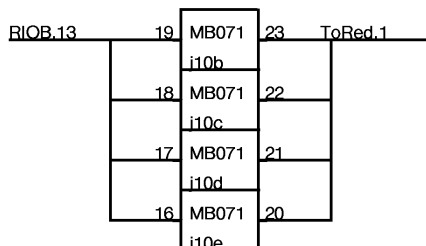
WriteMixerLo'Dc 2



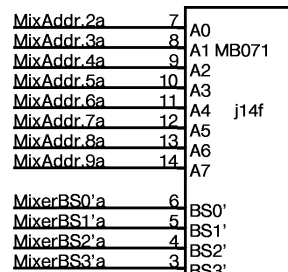
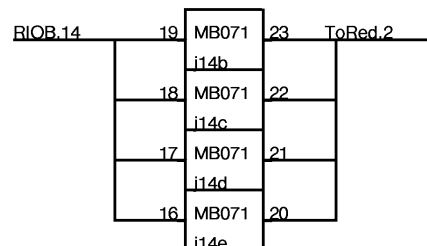
WriteMixerLo'Db 2



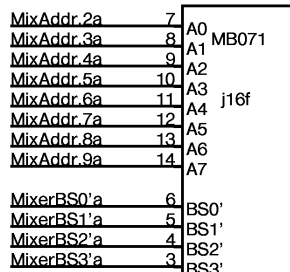
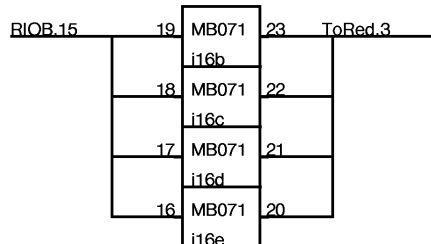
WriteMixerHi'Bc 2



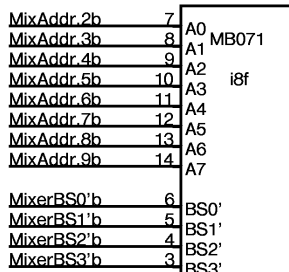
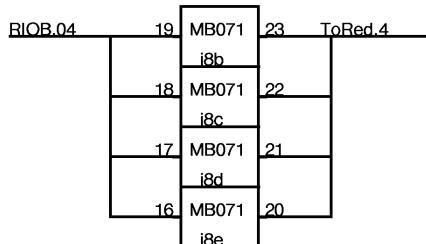
WriteMixerHi'Bc 2



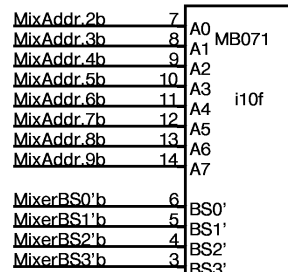
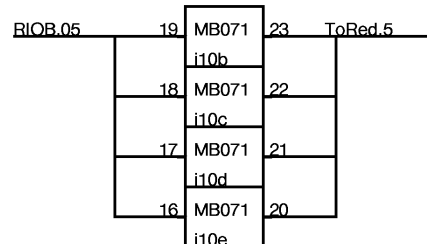
WriteMixerHi'Dc 2



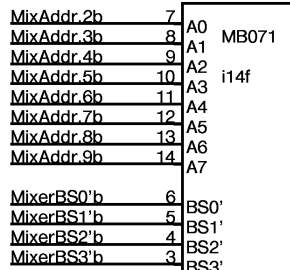
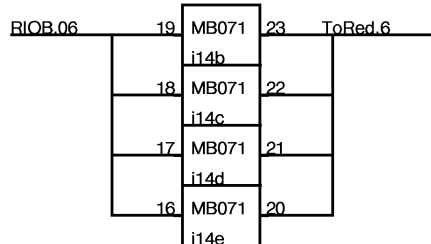
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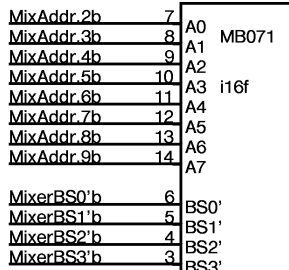
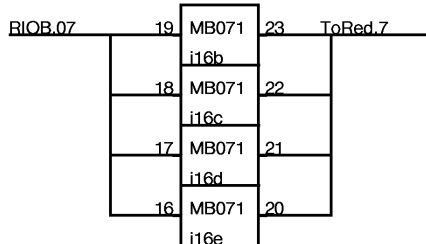
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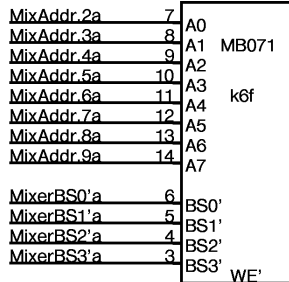
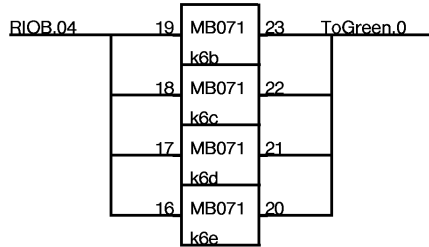
WriteMixerLo'Ba 2



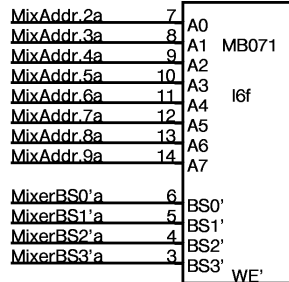
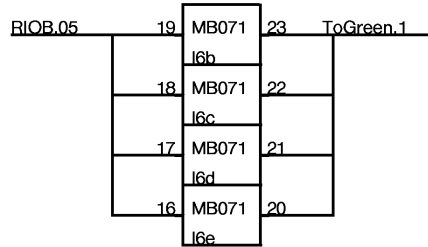
WriteMixerLo'Da 2



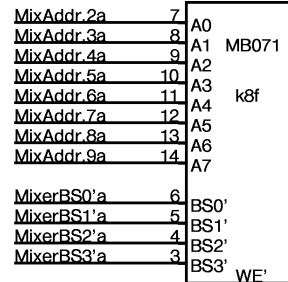
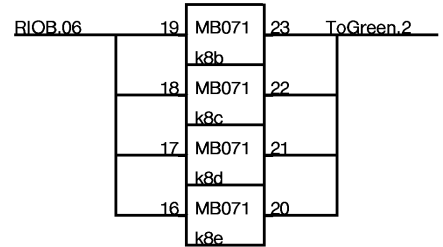
WriteMixerLo'Da 2



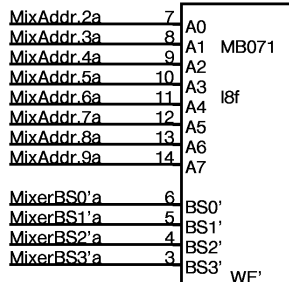
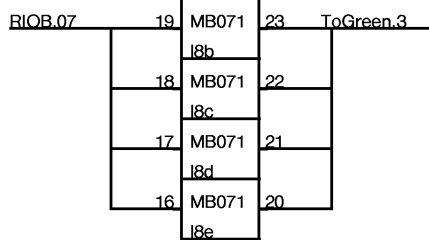
WriteMixerHi'Ba 2



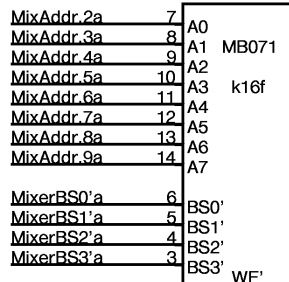
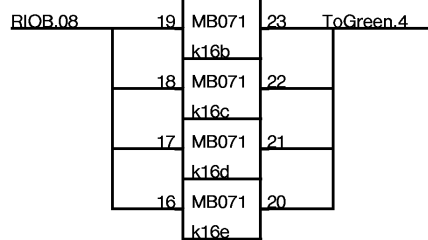
WriteMixerHi'Bb 2



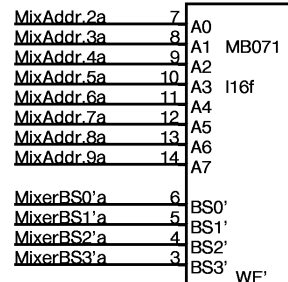
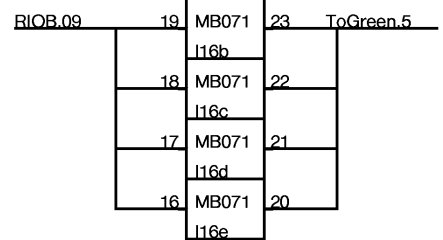
WriteMixerHi'Ba 2



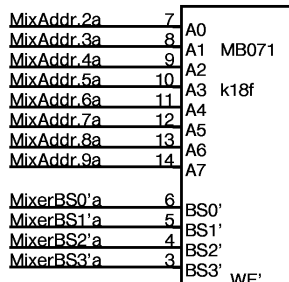
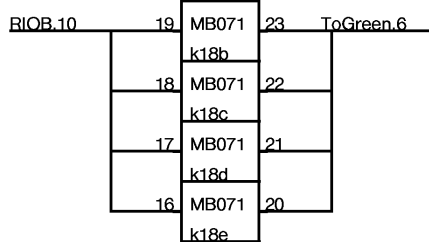
WriteMixerHi'Bb 2



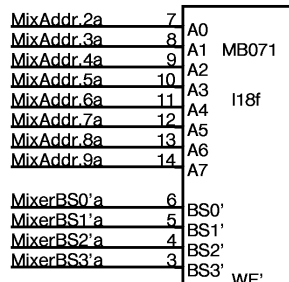
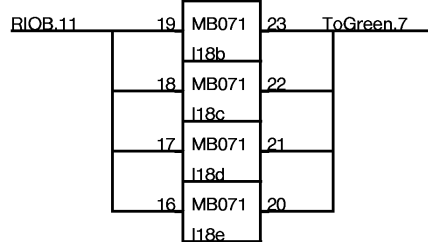
WriteMixerHi'Da 2



WriteMixerHi'Da 2

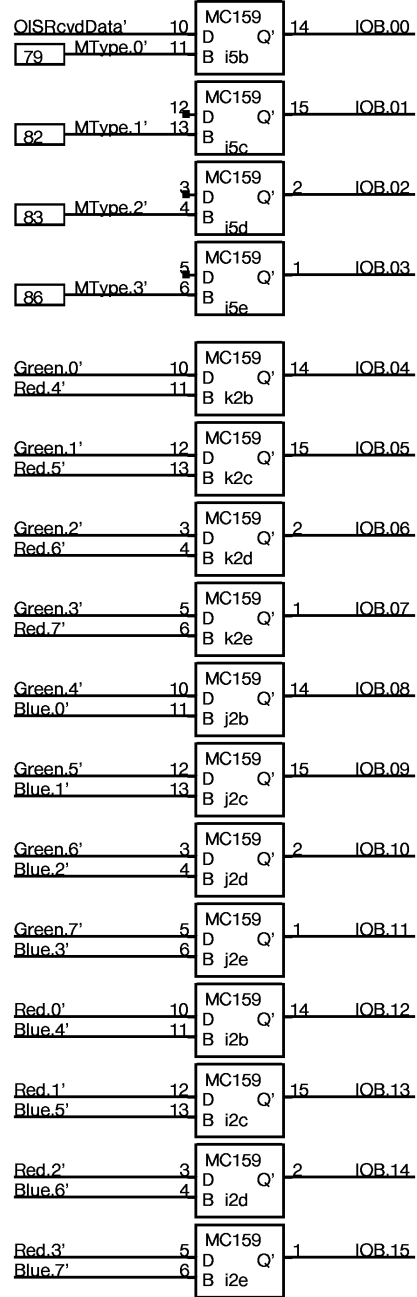
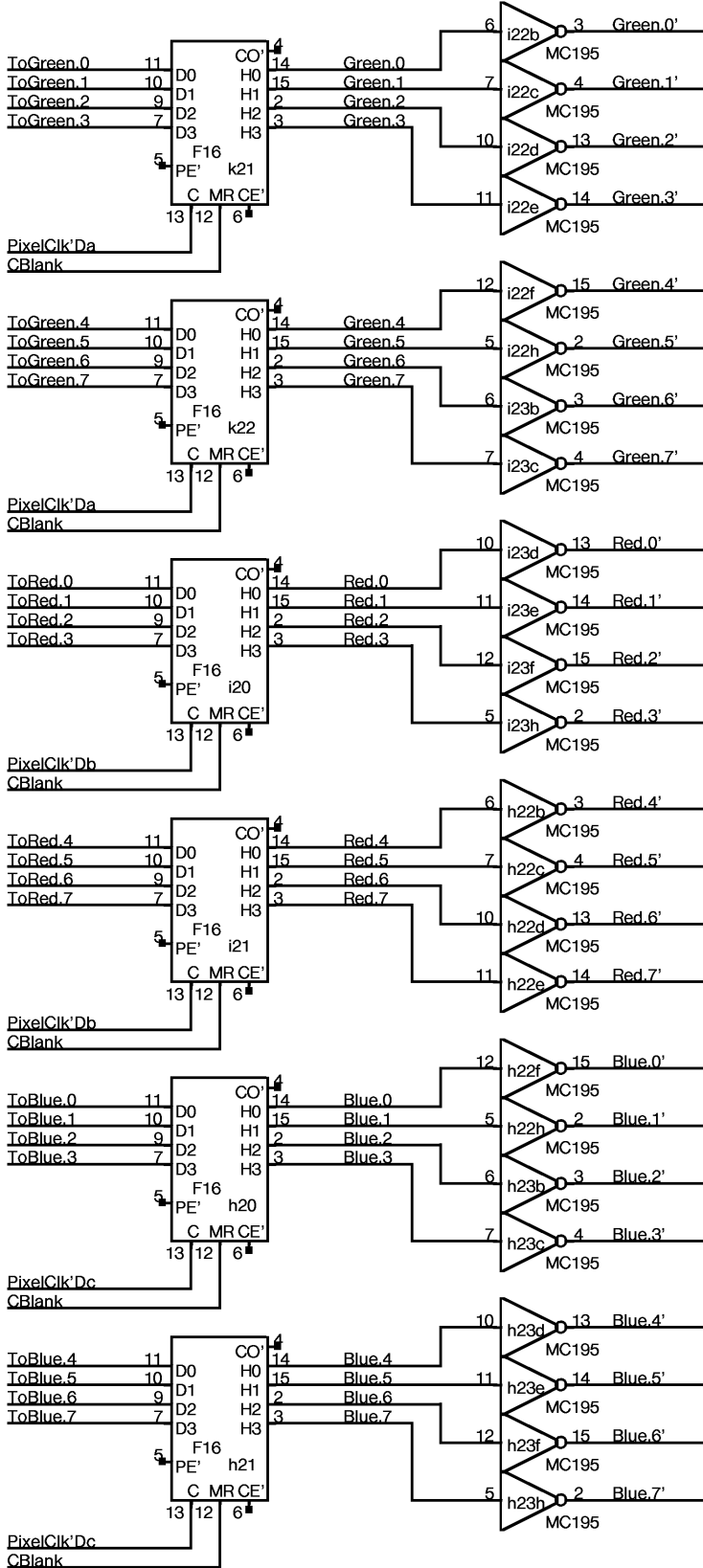


WriteMixerHi'Db 2



WriteMixerHi'Db 2

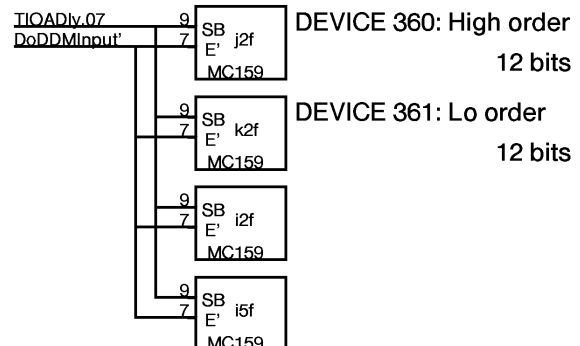
No Parity supplied
Use InputNoPE



Device 360:
keyboard data

Device 360:
Here I Am!!

Device 361:
Monitor type code
Jumped on
backplane

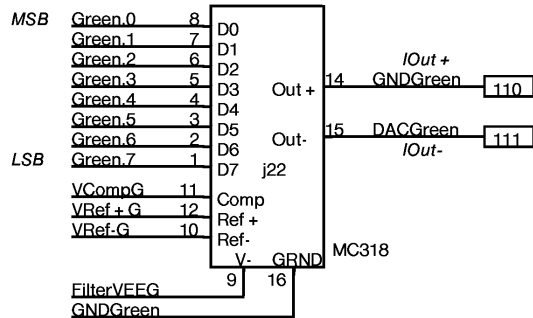


DEVICE 360: High order
12 bits

DEVICE 361: Lo order
12 bits

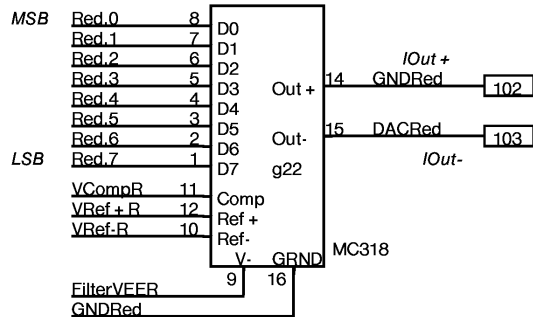


NOTES:
GNDGreen is
used as single point GND
for DAC system



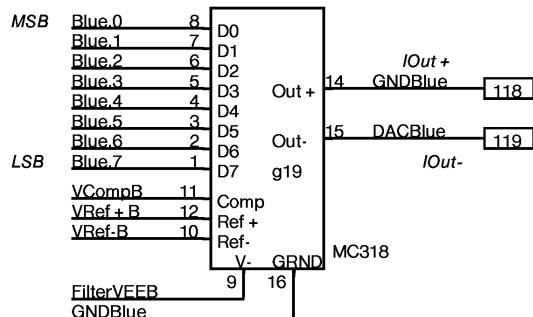
Digital/Analog converter

NOTES:
GND RED is
used as single point GND
for DAC system

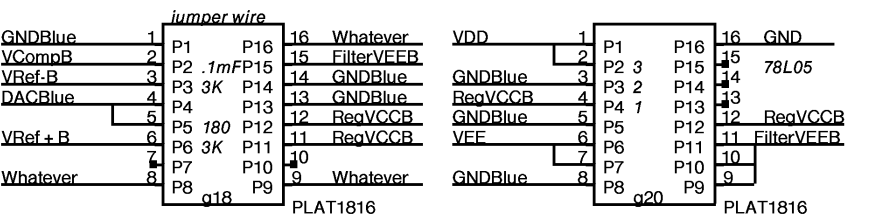
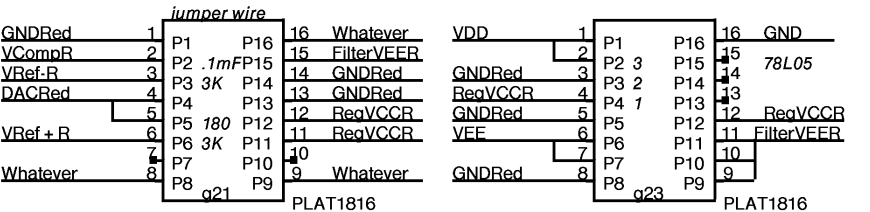
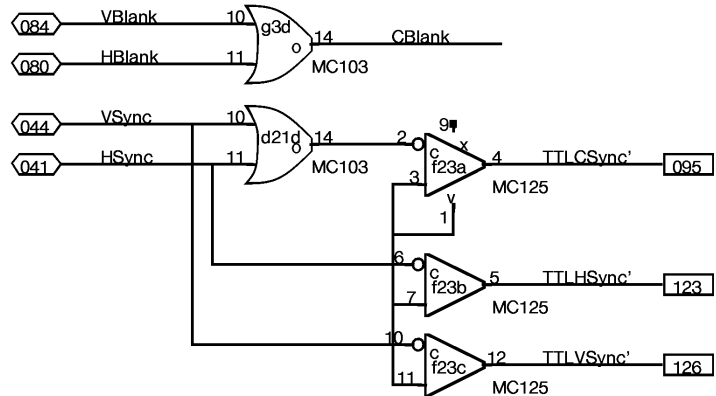
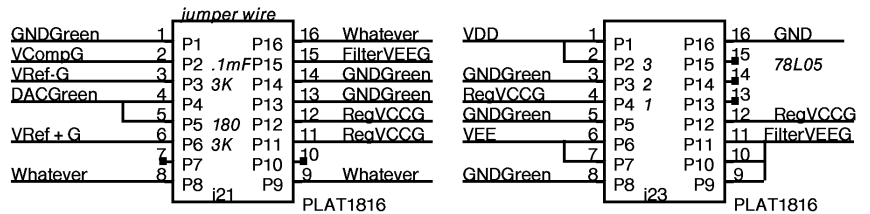


Digital/Analog converter

NOTES:
GNDBLue is
used as single point GND
for DAC system



Digital/Analog converter

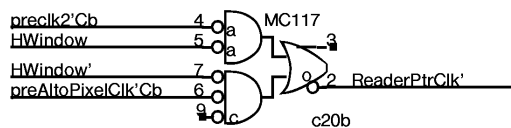
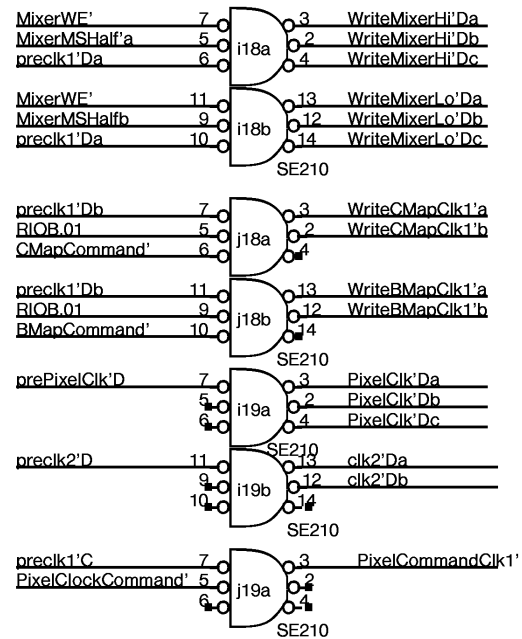
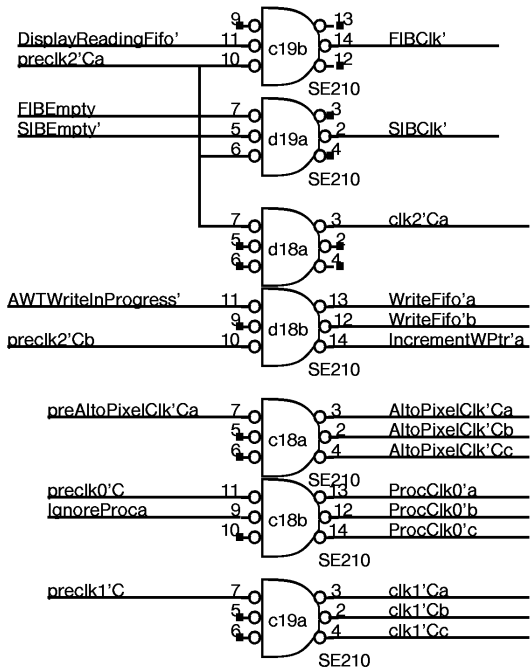
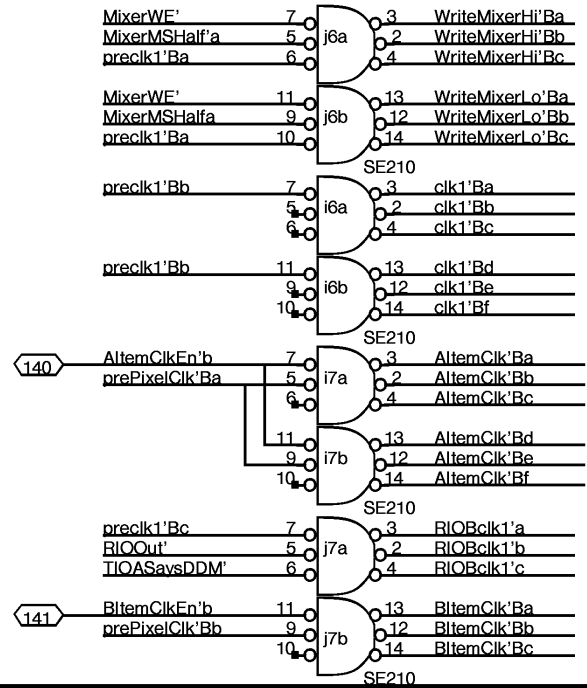
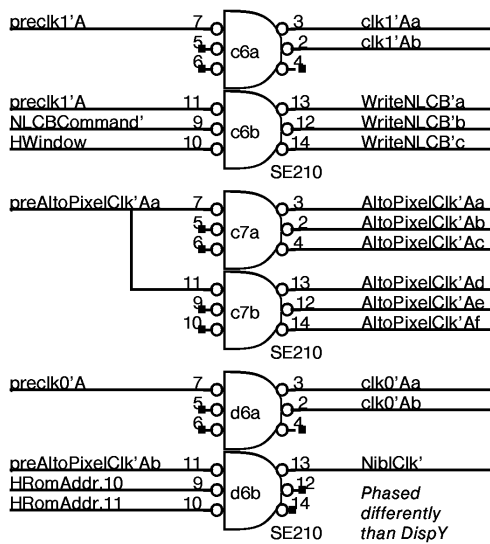


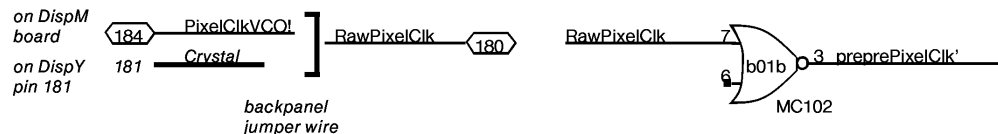
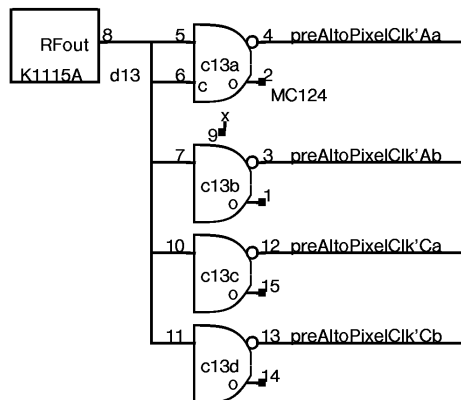
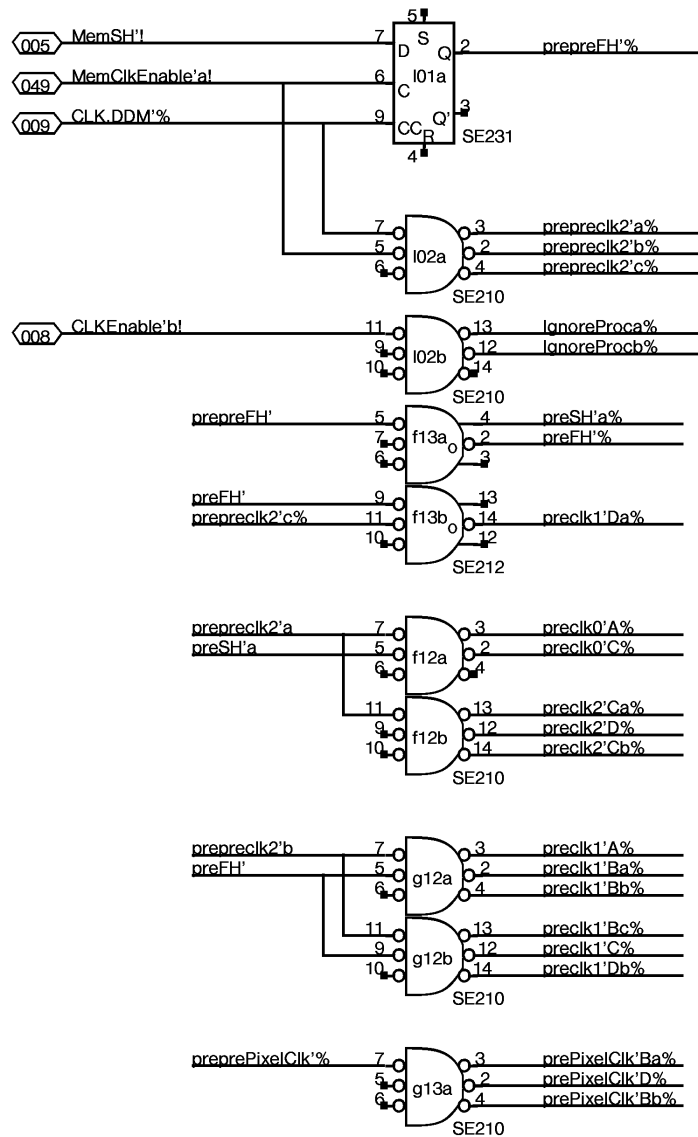
1	jumper wire	16
2	0.1mFarad	15
3	3 KOhm	14
4		13
5	180 Ohm	12
6	3 KOhm	11
7		10
8		09

1	0.1mFarad	16
2	in ³	78L05
3	gnd ²	14
4	out ¹	13
5	0.1mFarad	12
6	12 mHnry	11
7	2 ohm	10
8	+ 22 mF	09

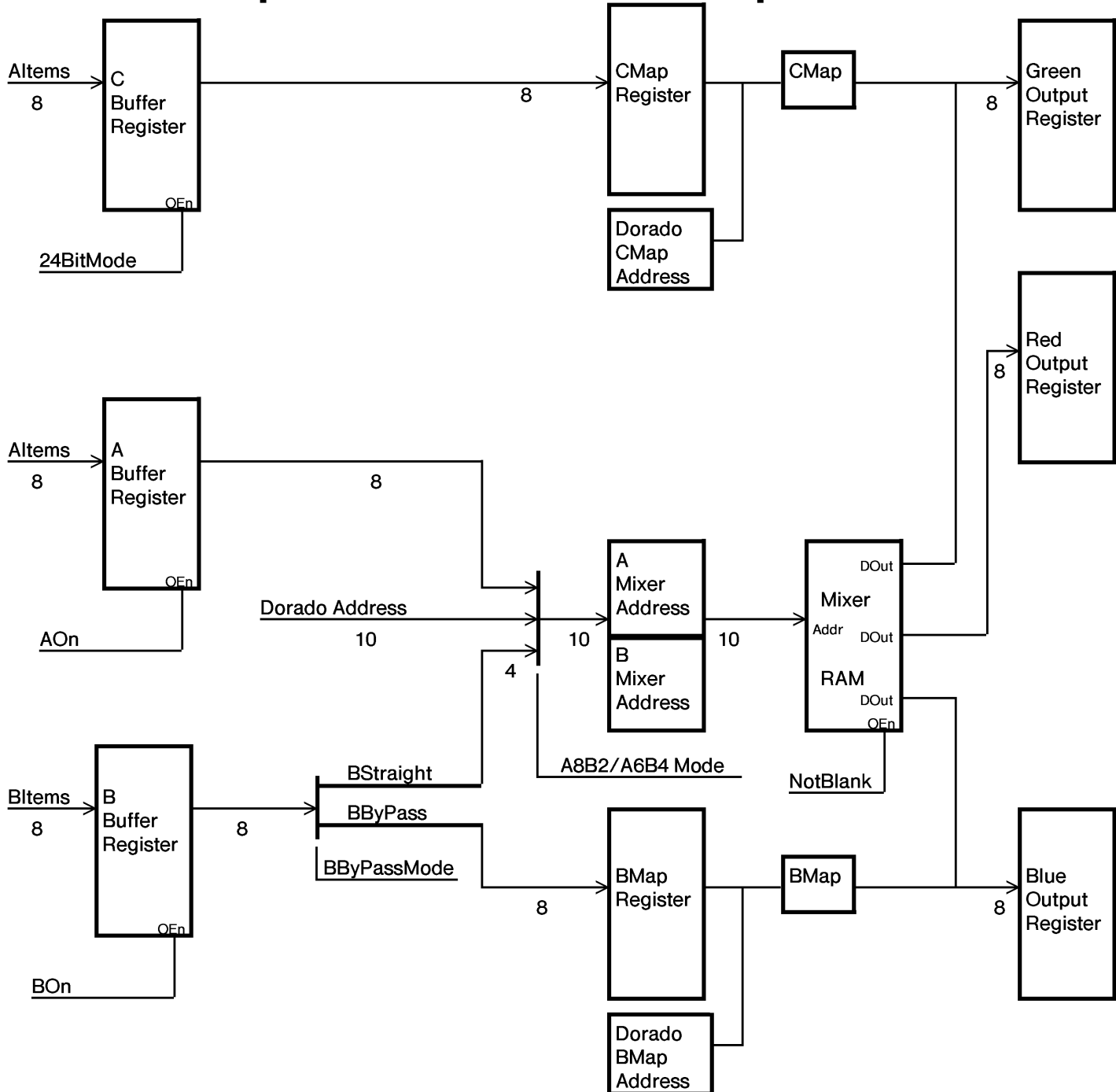
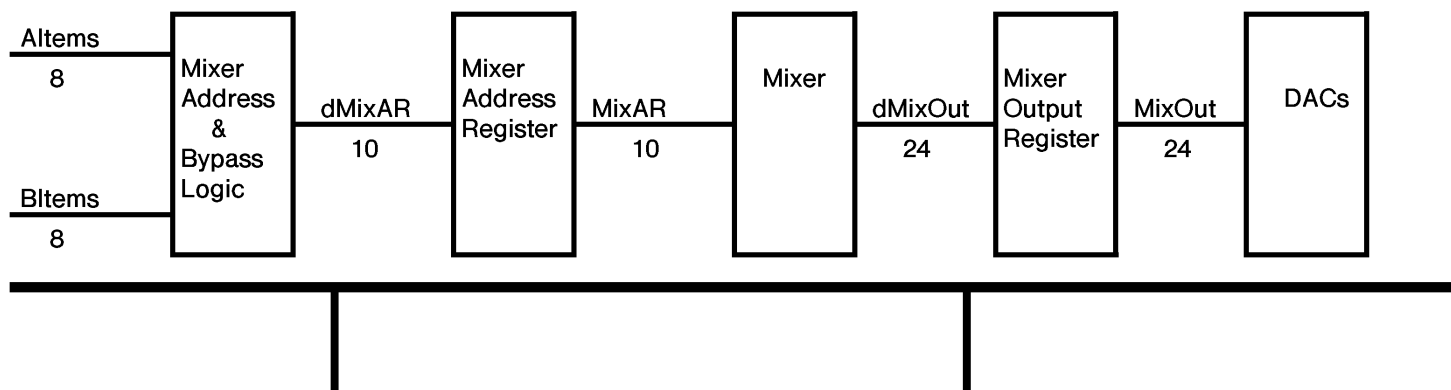
locations g18,g21,j21

locations j23,g23,g20





CONNECTOR		CONNECTOR		TIOA		TWI		CONNECTOR		IOBLo		CONNECTOR		IOBHi		CLK		C					
A	B	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r				
1	41	DISC VCO 23	PCLK 8,25,23,23 102		TIOA 176 21	TIOA 176 21	BBuf F16 12	ABuf F16 12	ABuf F16 12	RIOB 176 21	RIOB 176 21	RIOB 176 21	25 SE231						1				
2		DISC VCO 23	12040 VCO		WakeAHT 135	TIOA = D 113	BBuf F16 12	CBuf F16 12	CBuf F16 12	DIOB 159 20	DIOB 159 20	DIOB 159 20	25 SE210						2				
3	42	3140 VCO 23	F16 VCO	DISC 23	OISDrvr 10,4,10 105	TIOA = D A,21,21,10 113	WakeAWT AVSync 231	MapLd 13,14,21, 103 22	MixLd 16,16,20, 103 16	Modes B,12,12 195 12,15,2		DHM 105 12	DHMMix 231 16,B						3				
4		DISC VCO 23	95016 VCO	VCO 23	OISDrvr 101 10		BMapAd F16 12	ABufEn 231 12,B	ORs 12,16,16, 103 16	13,14,12,D 103		DHMap 231 13,14							4				
5	43	1648 VCO 23	95016 VCO	K1115 VCO	LS109 VCO		BMapAd F16 12	CMapAd F16 12	DCMapAd F16 14	IOB 159 20		CMapAd F16 12	DCMapAd F16 14						5				
6		DISC VCO 23	95016 VCO	Clock 210 24	Clock 210 24	ShutUp 231 4	DBMapAd F16 13	Mixer B0 17	Mixer B1 17	Clock 210 24	Clock 210 24	Mixer G0 19	Mixer G1 19						6				
7	44	AltoStoP 141 10	Nibble 176 10	Clock 210 24		CLCBDCd 161 5	DBMapAd F16 13	--	--	Clock 210 24	Clock 210 24	--	--						7				
8		CursorX F16 9	OIS 174 10	OIS 9,9,10,10 103	Width F16 4	LMarg F16 4	NLCB 145A 5	Mixer B2 17	Mixer B3 17	Mixer R4 18	Mixer R0 18	Mixer G2 19	Mixer G3 19						8				
9	45	CursorX F16 9	CursorHi 141 9	CursorLo 141 9	Width F16 4	LMarg F16 4	NLCB 145A 5	--	--	--	--	--	--						9				
10		CursorX F16 9	CursorHi 141 9	CursorLo 141 9	Width F16 4	LMarg F16 4	NLCB 145A 5	MixAddr F16 15	MixAddr F16 15	Mixer R5 18	Mixer R1 18	MixAddr F16 15	MixAddr F16 15						10				
11	46	VCW F16 4	Cursor 104 9	Width 4,4,6,4 103	LMarg 103 4	NLCBAddr F16 5	NLCBAddr 197 5	MixAddr F16 15	MixAddr F16 15	--	--	MixAddr F16 15	MixAddr F16 15						11				
12		PixelClk 10,10,6 176 10,F,6	OISSkew 176 10	3,4,4,2 104	HWindow 135 5	LoadASR 121 3	Clock 210 25	Clock 210 25	MixAddr F16 15	BMap 13	BMap 13	MixAddr F16 15	MixAddr F16 15						12				
13	47	HRomAddr 6	HRomOut 176 6	TTLtoEcl 25	Xtal K1115 25	LoadASR F16 3	Clock 212 25	Clock 210 25	MixAddr F16 15	--	--	MixAddr F16 15	BSel 171 15						13				
14		HRomAddr 6	ASR 141 3	ASR 141 3	ASR 141 3	ASR 141 3		Mixer B4 17		Mixer R6 18	Mixer R2 18	CMap 176 21	CMap 176 21						14				
15	48	HRomAddr 6		FIB 176 3	FIB 176 3	FIB 176 3		--		--	--	--	--						15				
16		HRom MC149 6	Fifo 3	Fifo 3	Fifo 3	Fifo 3	FifoAd 158 2	Mixer B6 17	Mixer B5 17	Mixer R7 18	Mixer R3 18	Mixer G4 19	Mixer G5 19						16				
17	49		--	--	--	--	FifoAd 158 2	--	--	--	--	--	--						17				
18		6,23,4,2 102	OISRcvd 231 10,B	Clock 210 24	Clock 210 24	RPtr 2	WPTr 2	BIDac Plat 22	Mixer B7 17	Clock 210 24	Clock 210 24	Mixer G6 19	Mixer G7 19						18				
19	50	6,4,6,6 103	6,4,6 9,4,6 195	Clock 210 24	Clock 210 24,B	RPtr 2	WPTr 2	BIDac 10318	--	Clock 210 24	Clock 210 24	--	--						19				
20			HRom MC149 6	RdrPtrClk CTisAWT 117 24,8	ASRSync 231 2		AOn ASRSync 135 2,4	BIDac Plat 22	Blue 20	Red 20	PXLCLK 176 21	176 21	176 21						20				
21	51			SCAN F16 4	7,2,2,22 103	Flags 231 7	Flags 118 7	RdDac Plat 22	Blue 20	Red 20	GrDac Plat 22	Green 20							21				
22		SIB 176 3	SIB 176 3	SIB 176 3		CMND 176 21	CMND 176 21	RdDac 10318	Color' 195 20	Color' 195 20	GrDac 10318	Green 20							22				
23	52	8,8,C 105	ProcClk0 176 8		blocked 135 8	ASRSync 135 2	TTLCSync 22 125	RdDac Plat 22	Color' 195 20	Color' 195 20	GrDac Plat 22		1,B,C,D 102						23				
24		NextAWT 113 8			WakeCnt B,4*8,2 195	WakeCnt F16 8	Hold 231 8	IOInOut 1660 21	TIOA = D 161 21	Fout 176 1	Fout 176 1	Fout 176 1	FtTsk 113 1						24				
C		a	b	c	d	e	f	g	h	i	j	k	l	D									
E		11	26	39	55	70	86	99	114	129	143	159	174	E									
		NEXT	SubT	BLK	FIN	Hold	IOIn/out	CONNECTOR	IOB FNXT	FOUT	FTsk	DMux											
XEROX		Project	Reference	File		Designer		Rev	Date	Page													
PARC		Dorado	DispM Board Layout	DispM26.sil		K. Pier		Ch	11/09/82	26													



DispY	SIZE	DispM
Altem	8	xxx
Bltem	8	xxx
CursorData	1	117
AltemClkEn	1	140
BltemClkEn	1	141
AOff	1	048
BOff	1	045
HSync	1	041
HBlank	1	080
HalfLine	1	081
VSync	1	044
VBlank	1	084
RawPixelClk	1	184
Crystal	1	181
PixelClkVCO	1	180
Modes	3	xxx
XHSync	1	036
XVSync	1	037
XSyncEn	1	040
ABypass	NC 1 NC	029
		RefIn

85-88-89-92-93-96-97-100
101-104-105-108-109-112-113-116
28-32-33

Red GND	1	102
Red	1	103
Green GND	1	110
Green	1	111
Blue GND	1	118
Blue	1	119
TTLCSync'	1	095
TTLCSync'GND	1	096

DispM to coax/BNC connectors

MType.0	1	079
MType.1	1	082
MType.2	1	083

Monitor type field
Jumper resistors on backplane

DispM	ControlA
121	WakeTHT (TWReg4) twisted pair 56
120	WakeTWT (TWReg9) twisted pair 128

Plus seven wire interface cable. See page 11.

DDC Slow IO System

DEVICE	TIOA	I/O	TASK	FORMAT and COMMENTS
AStatics	367	O	AHT	
NLCB	366	O	AHT	
BMap	365	O	CHT, EMU	Keep, Write, LoadAddr, 0, DataOrAddr8-15
AHTFlag	364	O	AHT	
AWT	363	O	AWT	
CMap	362	O	CHT, EMU	Keep, Write, LoadAddr, 0, DataOrAddr8-15
MIXER	361	O	CHT, EMU	Keep, Write, LoadAddr, 0, Data.4-15
PIXELCLK	360	O	CHT, EMU	Pixel clock rate
STATUS	361	I	CHT, EMU	MType.0-3, Green.0-7, Red.0-3
STATUS	360	I	CHT, EMU	Keyboard, 1, 1, 1, Red.4-7, Blue.0-7

00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	
				Pixel Clock Rate								Clock Divider				PIXELCLK
Keep Mixer'	Write Mixer'	Load Mixer Addr	X		Addr.0	Addr.1	Addr.2	Addr.3	Addr.4	Addr.5	Addr.6	Addr.7	Addr.8	Addr.9	Hi/Lo select	MIXER
				Mixer Data 12 bits												
Keep BMap'	Write BMap'	Load Bmap Addr	X					Address.0-7 OR Data.0-7								BMap
Keep CMap'	Write CMap'	Load CMap Addr	X					Address.0-7 OR Data.0-7								CMap
														AWT Shut Up	AHT Shut Up	AStatics
NLCB Addr 00	NLCB Addr 01	NLCB Addr 02	NLCB Addr 03	NLCB DATA 12 Bits												NLCB
											IOFetch				Set/Clr Cur WCB Flag	AWT
														X	Must Be 0	AHTFlag

1	jumper wire	16
2	0.1mFarad	15
3	3 KOhm	14
4	120 Ohm	13
5	180 Ohm	12
6	3 KOhm	11
7		10
8		09

locations

g18,g21,j21

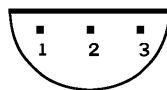
3 identical copies

1	0.1mFarad	16
2	in ³ 78L05	15
3	gnd ²	14
4	out ¹	13
5	0.1mFarad	12
6	12 mHnry	11
7	2 ohm	10
8	+ 22 mF	09

locations

j23,g23,g20

3 identical copies



1. Output
2. Ground
3. Input

Bottom view of pinout
for 78L05

BOTTOM VIEW!!
BOTTOM VIEW!!

SIP in location g41 is 100 ohm terminator with leg 6 cut (making DDMTIOA = 360B)

SIP in location b52 is 100 ohm terminator with legs 3 and 4 cut for Task 9D = 11B

SIPs in locations d42 and e42 are 220 ohm value instead of 100 ohm (terminators for 7 wire interface lines)

Crystal oscillators, type K1115A:
location c5, value 10 MHz, for VCO
location d13, value 20 Mhz for Alto
value 50 MHz for LF

Horizontal PROM, type MC149
location a16 and b20 for LF display
location b20 ONLY for Alto style display
programmed for each monitor type

1	8.2 KOhm	16
2	Bead	15
3	4700 pF	14
4	22 microF +	13
5	8.2 KOhm	12
6	+ 22 microF	11
7	4700 pF	10
8	Bead	09

location a1

1		16
2	2.2 KOhm	15
3	2.2 KOhm	14
4	2200pF	13
5	2200pF	12
6	2.2 KOhm	11
7	2.2 KOhm	10
8		09

location a2

1		16
2	4.7 KOhm	15
3	+ MV1403-	14
4	+ MV1403-	13
5	180 nH	12
6	15 pF	11
7	1 KOhm	10
8		09

location a4

1	jumper wire	16
2	0.1microF	15
3	0.1microF	14
4	+ 22 microF13	13
5	1 KOhm	12
6	Bead	11
7		10
8		09

location a6

1		16
2	jumper wire	15
3		14
4		13
5		12
6		11
7		10
8		09

location c3

Rev.	Date	Page	Revisions
Ch	11/07/82	7, 21, 23, 26 30 31	Expanded pixel clock buffer. Added 120 ohm resistor to position 4 of Platform for g18, g21, and j21. Added Revision Record Page